What carries the gains of debiased graph recommendation? A controlled study on four benchmarks with unbiased test data

Anonymous

Abstract

Exposure-bias corrections for graph recommenders are usually compared on benchmarks with uniformly exposed test data, but their protocols, tuning budgets and selection rules differ. We re-evaluate these corrections on Coat, Yahoo! R3, KuaiRand-Pure and KuaiRec under one protocol. Uniform data are used for validation and testing, with one propensity exception on Coat that we check for the prior rows only. Configurations are selected per split on validation data, and the selected configurations of the leading methods are retrained with three further seeds. Selecting on test data inflates Coat scores by up to 0.030 nDCG@5 and, under the per-split test oracle, reverses the order of iALS and LightGCN. Against an equally tuned LightGCN backbone, third-party graph debiasers give point-estimate gains of at most +4.10 ($\times 10^{-3}$) nDCG on Coat, Yahoo! R3 and KuaiRand-Pure (+5.26 in the seed 1 to 3 re-runs). The Coat 95% half-widths of these debiasers reach ± 30, and none of the gains survives a correction for overlapping splits. Our own QP-LightGCN family adds +16.77 on Yahoo! R3 and +3.93 on KuaiRand-Pure over LightGCN, and it is not significant after that correction. An additive prior fitted on the exposed log and added as a score offset gives the largest point-estimate gains over LightGCN on Coat and KuaiRec, and none on Yahoo! R3 or KuaiRand-Pure. Removing its inverse-propensity weights does not reduce the gain, a shuffled item term removes the KuaiRec gain, and popularity does not reproduce it. The Coat gain is not robust to the correction for overlapping splits. Making the multi-hop propagation exactly unbiased does not improve ranking on Coat, Yahoo! R3 or KuaiRec; DR and DRUP were not run on KuaiRand-Pure. We propose no new method. We do not claim that debiasing is ineffective in general, and we do not claim equivalence on Coat, which is underpowered.

Keywords: recommender systems, exposure bias, graph collaborative filtering, propensity, doubly robust

1. Introduction

Graph-based recommenders represent users and items as nodes and score a user-item pair by propagating embeddings over the user-item graph. LightGCN (He et al., 2020) is a standard backbone of this family, and it is usually trained on implicit feedback logs. In such a log an interaction is recorded only when the item was shown and the user responded. The logged graph therefore reflects the exposure policy as well as user taste (Marlin and Zemel, 2009; Chen et al., 2023). A model fitted to the log treats every absence as informative, and the resulting bias is the subject of a large literature on data missing not at random.

The pairwise loss that trains LightGCN, Bayesian personalised ranking (Rendle et al., 2009), draws its negatives uniformly from the items the user has no logged interaction with. Under exposure bias most of these items were never shown, and some of them would have been liked had they been shown. Such false negatives may be concentrated among rarely exposed items, where the log carries the least information. Matrix factorisation models address exposure by treating it as a latent variable (Liang et al., 2016) or by reweighting the likelihood with propensities (Saito et al., 2020). Pointwise inverse propensity and doubly robust (DR) estimators (Schnabel et al., 2016; Wang et al., 2019b; Guo et al., 2021) act on the loss. Graph models have received their own corrections, among them inverse-propensity aggregation (Kim et al., 2022), adjacency normalisation (Zhao et al., 2022) and post-hoc popularity correction (Islam et al., 2026b).

This paper starts from the two nuisance functions that a DR estimator already fits: an exposure propensity p_{ui} and an imputed preference $\hat{Y}_{ui}$. Under an independence assumption between exposure and preference given covariates, Bayes' rule gives the probability that a pair absent from the log is a positive,

$$Q_{ui} = \Pr\big(Y_{ui} = 1 \mid O_{ui}Y_{ui} = 0,\, x_{ui}\big) = \frac{\hat{Y}_{ui}\,(1 - p_{ui})}{1 - p_{ui}\,\hat{Y}_{ui}}. \tag{1}$$

The candidate models use Q in two places inside an otherwise unchanged LightGCN. Each sampled BPR negative is down-weighted by $1 - Q_{uj}$, and

the standardised logit of the imputation is added to the score as a fixed offset whose weight is chosen on validation data. Both nuisance functions are fitted on the biased log before training. The one exception to the rule that training uses no uniformly collected data is Coat, whose shipped propensities were estimated by the dataset authors from a uniform sample. We call the full model QP-LightGCN and evaluate it together with its variants as one family.

The gains reported for these corrections are hard to compare, for three reasons. First, the studies use uniformly collected data in different ways. Some train on it, some select configurations on it, and some use it only for testing (Section 2). Second, tuning budgets differ, so a method given a larger grid can obtain a higher test score through selection alone. Third, single runs are common, and the ordering of close methods can depend on the random seed. Our own runs illustrate the third point. In the main run, QP-LightGCN is above iALS on Coat by $+20.02 \times 10^{-3}$ [19.74], which is significant after Holm correction over the eight baseline comparisons ($p_{\text{Holm}} = 0.047$). When the selected configurations are retrained with seeds 1 to 3, the difference is $+3.39 \times 10^{-3}$ [20.70], with $p_{\text{Holm}} = 0.75$ over the two baseline comparisons of that check. These seed comparisons are descriptive, because the two baselines of each dataset were chosen on test data.

This study asks four research questions.

RQ1 How much do evaluation choices change the conclusions? We vary the selection split, the tuning budget, the retraining seed, the early-stopping rule and the multiple-comparison family.

RQ2 How much do graph-specific corrections add over an equally tuned backbone?

RQ3 Which ingredient carries the gains that remain?

RQ4 Does making the multi-hop propagation exactly unbiased matter for ranking?

Our reading of the evidence is as follows. On Coat, Yahoo! R3 and KuaiRand-Pure, once uniform data are kept out of training and configurations are selected on validation data, the third-party graph-specific corrections change the nDCG of an equally tuned LightGCN by point-estimate amounts of at most $+4.10 \times 10^{-3}$ (a few thousandths; $+5.26 \times 10^{-3}$ in the seed 1 to 3 re-runs). On Yahoo! R3 the inflation that selecting on the

test split adds to LightGCN (about 0.004 nDCG@5) is as large as the gain of the best of these corrections ($+4.10 \times 10^{-3}$ [3.29]). On Yahoo! R3 and KuaiRand-Pure our own QP-LightGCN family adds $+16.77 \times 10^{-3}$ [5.82] over LightGCN on Yahoo! R3 and $+3.93 \times 10^{-3}$ [1.53] on KuaiRand-Pure, but neither difference survives the correction for overlapping splits. The largest point-estimate effects in our runs come from two simple ingredients. The first is a fixed, inverse-propensity-fitted additive imputation added to the score, which we call the prior. It raises LightGCN on Coat ($+31.98 \times 10^{-3}$ [20.65]; this gain is not robust to the correction for overlapping splits) and on KuaiRec ($+21.22 \times 10^{-3}$ [2.86]), but not on Yahoo! R3 (-1.43×10^{-3} [3.93]) or KuaiRand-Pure (-0.81×10^{-3} [1.33]). Its inverse-propensity weights do not appear to be what matters in our runs: on KuaiRec, the iALS model with the same imputation without those weights scores $+8.05 \times 10^{-3}$ [2.06] above the one with them. The second is a training-free doubly robust propagation of the log, which has the highest mean of the methods we ran on the fully observed KuaiRec. Making that propagation exactly unbiased does not change ranking accuracy on the three real logs on which it was run (Coat, Yahoo! R3 and KuaiRec). On KuaiRec, DR-LightGCN lies $+13.17 \times 10^{-3}$ [3.31] above the BPR backbone, and the same backbone trained pointwise, without any debiasing, lies $+19.59 \times 10^{-3}$ [4.55] above it. The pointwise difference is larger, but the two differences are not compared by a paired test, so this is a difference of means.

The paper makes four contributions. None of them is a new estimator.

1. *A controlled protocol and its sensitivity (RQ1).* Four public benchmarks with unbiased test data (Coat, Yahoo! R3, KuaiRand-Pure and KuaiRec), uniform data used only for validation and testing, per-split selection on validation data, per-user paired tests with Holm correction, retraining with seeds 1 to 3, and configuration counts for every method. The new part is the sensitivity analysis. Selecting on the test split instead of on validation raises the score of the trained methods on Coat by 0.021 nDCG@5 on average (range 0.010 to 0.030).
2. *Graph-specific debiasing adds little over an equally tuned backbone (RQ2).* On Coat, Yahoo! R3 and KuaiRand-Pure, the third-party graph debiasers change the paired score of LightGCN by at most $+4.10 \times 10^{-3}$ [3.29], for r-AdjNorm on Yahoo! R3 ($p_{\text{Holm}} = 0.029$ in the primary family of that dataset, and not significant under the correction for overlapping splits). The bound applies to these third-party debiasers

only. The refinements of the prior that we built are reported as members of the evaluated family, with their configuration counts, and the QP-LightGCN family is the largest of them.

3. *Prior anatomy (RQ3).* The gains over the backbone that the prior provides are examined with controls. Removing the inverse-propensity weights does not reduce them, a shuffled item term removes the KuaiRec gain, and popularity does not reproduce it. The shuffled and popularity weightings of the refinements are reported as controls. The mechanism is not identified.
4. *A negative result on walk correction (RQ4, secondary).* The walk correction of DRUP removes the repeated-edge bias of multi-hop DR propagation. On KuaiRec, DRUP minus DR is -0.12×10^{-3} [0.15], which is not significant ($p_{\text{Holm}} = 0.113$). On Coat, at matched configurations, the correction lowers the mean nDCG@5 for every value tested.

We do not propose a new estimator, and the refinements in Section 4 are members of an evaluated family, not contributions in their own right. We do not claim that debiasing is useless in general. The scope is four public benchmarks, offline evaluation, top-K nDCG at the cut-off used for each dataset (5 for Coat and Yahoo! R3, 10 for KuaiRand-Pure, 20 for KuaiRec), and implicit-feedback learners trained on the biased log. Coat has 225 evaluated users, so its power is low. For Coat we report the differences that the data exclude, and we do not report equality. We do not claim that AdvInfoNCE, NAVIP, r-AdjNorm, SimGCL or MACR fail in general. Our implementations, tuned with the stated budgets under this protocol, do not beat the backbone by more than the amounts reported. Propagation with DR weights is not generally the best method either: on Yahoo! R3 its mean score is below that of trained LightGCN. We do not give a mechanism for why the prior helps beyond the controls reported in Section 5.3. The reading that the prior acts as a debiased item-quality score is a hypothesis.

Section 2 reviews related work and positions the study. Section 3 describes the protocol: datasets, leakage controls, selection, seeds and statistical tests. Section 4 describes the evaluated methods, including the prior family. Section 5 reports the results for RQ1 to RQ4. Section 6 discusses why the effects are small, gives recommendations for evaluating debiasing methods and states the limitations. Section 7 concludes.

2. Related work

2.1. Graph collaborative filtering and graph debiasing

LightGCN (He et al., 2020) is the backbone used in our experiments. Other graph convolutional recommenders include NGCF (Wang et al., 2019a) and UltraGCN (Mao et al., 2021), and contrastive graph variants include SGL (Wu et al., 2021) and SimGCL (Yu et al., 2022). Linear and closed-form models include EASE (Steck, 2019), GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023). Weighted matrix factorisation for implicit feedback, iALS (Hu et al., 2008), is a standard baseline, and Rendle et al. (2022) revisit its tuning on item recommendation benchmarks. Wu et al. (2023) survey graph neural recommenders. These models fit or propagate over the logged graph and do not model which pairs were exposed.

Graph models have also been debiased. NAVIP (Kim et al., 2022) weights neighbour aggregation by inverse propensities. r-AdjNorm (Zhao et al., 2022) tunes the strength of adjacency normalisation to trade accuracy against novelty, and Chen et al. (2024) study how graph convolutions can amplify popularity bias. APDA (Zhou et al., 2023) and CAGED (Que et al., 2025) learn aggregation weights, and DPAA (Islam et al., 2026a), an arXiv preprint, reweights message passing. AdvDrop (Zhang et al., 2024) learns an adversarial edge dropout on the interaction graph and reports IPS and DR baselines on Coat, Yahoo! R3 and KuaiRec. PPD (Islam et al., 2026b) removes a popularity direction from pre-trained LightGCN embeddings without retraining, and is evaluated on KuaiRec, Coat and Yahoo! R3. DR-GNN (Wang et al., 2024) applies distributionally robust optimisation over graph neighbourhoods; its "DR" stands for distributionally robust, not doubly robust. The popularity-debiasing methods MACR (Wei et al., 2021) and PDA (Zhang et al., 2021) are used as baselines in our study. Most of these methods act on aggregation, adjacency, embeddings or a robust objective, rather than on the negatives of the pairwise loss.

2.2. Propensity, doubly robust and negative-weighting methods

Ratings and implicit feedback are missing not at random (Marlin and Zemel, 2009), and surveys organise the resulting biases and the causal methods that address them (Chen et al., 2023; Gao et al., 2024). Propensity methods re-weight the loss by inverse propensities or self-normalised estimators (Schnabel et al., 2016; Swaminathan and Joachims, 2015; Saito et al., 2020). Steck (2010) showed that missing ratings carry information, proposed

test measures that can be estimated without bias when ratings are missing not at random, and derived training objectives that account for the missing entries. Doubly robust learning adds an imputation model to the inverse-propensity estimator (Wang et al., 2019b), and later doubly robust variants address related debiasing problems (Guo et al., 2021; Dai et al., 2022; Li et al., 2023). Information-theoretic counterfactual learning from missing-not-at-random feedback is studied by Wang et al. (2020).

Some debiasing methods use a small uniform sample for training. AutoDebias (Chen et al., 2021) learns sample weights and an imputation function by bi-level meta-learning on a small uniform sample. CausE (Bonner and Vasile, 2018) embeds a small randomised sample alongside the logged one, and KDCRec (Liu et al., 2020) distils from a model fitted on uniform data into a model trained on the biased log.

The methods closest to our negative weighting are those for exposure and false negatives. ExpoMF (Liang et al., 2016) infers an exposure posterior for unclicked items inside a matrix factorisation model. Rel-MF (Saito et al., 2020) re-weights observations by propensity, and unbiased pairwise learning (Saito, 2020) applies the same idea to pairwise objectives. SRNS (Ding et al., 2020) identifies likely false negatives during sampling, and debiased contrastive learning (Chuang et al., 2020) corrects negatives that are in fact positives. In contrastive collaborative filtering, the BC loss (Zhang et al., 2022) adds a margin that depends on the popularity-bias degree of each interaction, and AdvInfoNCE (Zhang et al., 2023) assigns each negative an adversarially learned hardness weight. Correct-and-Weight (Yin et al., 2026), an arXiv preprint, weights negatives against false negatives and is tested with graph backbones on random splits. Exposure-aware negative sampling for graph recommenders uses in-community popularity as a proxy for exposure (Chen et al., 2026), also an arXiv preprint. The original formulations of ExpoMF, SRNS, debiased contrastive learning and AdvInfoNCE do not combine a propensity with an imputation of the kind a doubly robust estimator provides. IPS-weighted BPR on LightGCN has been studied with simulated propensities (Raja and Vats, 2025).

The offset used by our candidate models is a fixed, standardised imputation logit added to the score. Adding a fixed offset is a familiar device. In long-tail classification, logit adjustment adds a class prior to the logits (Menon et al., 2021). The offset used here differs in one respect: it is a debiased estimate of item-level preference, obtained from an inverse-propensity-weighted imputation, and not a class prior.

2.3. Evaluation and reproducibility

Several studies have shown that comparisons between recommenders depend on the protocol. Ferrari Dacrema et al. (2019) re-ran recent neural recommenders against simple baselines and found that several did not outperform well-tuned versions of those baselines. Rendle et al. (2019) argue that baselines in published comparisons are often under-tuned, so that a reported gain depends on the effort spent on each method. Rendle et al. (2022) show that iALS, when tuned carefully, is competitive on common item recommendation benchmarks. Anelli et al. (2023) question the reported advances of graph collaborative filtering.

Several parts of the protocol can change a conclusion. Sampled and exact metrics can disagree (Krichene and Rendle, 2020). Nadeau and Bengio (2003) propose a corrected resampled t-test for the generalisation error, for resamples that overlap. Reporting the expected validation performance as a function of the number of configurations tried makes the tuning budget visible (Dodge et al., 2019). Comparisons over several methods and datasets also need a multiple-testing correction, for which we use the step-down procedure of Holm (1979).

2.4. Gap and positioning

The contribution of this paper is evidence rather than a new method. The graph debiasers and the candidate models of Section 4 are evaluated under one protocol, and the study measures how far each conclusion depends on evaluation choices. Table 1 summarises how the related studies use uniformly collected data and which benchmarks they use. The uses differ. Some studies train on uniform data (AutoDebias, CausE, ConAlign), some select configurations on part of it (CDR, PPD, ConAlign), and some keep it out of both training and selection (DCE, BISER).

A web search in October 2026 found no graph recommender that uses a posterior probability, built from a propensity and an imputation, to weight sampled negatives, and none that adds a debiased imputation to a graph model's score. It also found no graph study that uses KuaiRand-Pure. The search used general web sources and cannot be exhaustive. Coat, Yahoo! R3 and KuaiRec are already used for graph debiasing by AdvDrop, PPD and DPAA. We therefore do not claim that the posterior weighting or the imputation offset is new. The claim we make is narrower: we found no study that evaluates graph debiasers on all four benchmarks under one protocol,

with uniform data excluded from training, and with an analysis of how the budget and the selection rule change the conclusions.

3. Protocol

The protocol rests on one rule. Uniformly collected data are read only to choose among trained models and to measure them. They never enter a training objective, a nuisance model that a trained model uses, or the stopping of training. This section states the controls that enforce the rule, how the splits, the selection and the seeds are built, how the tests and sensitivity margins are defined, and how the stored results were reconciled before the analysis. The methods to which the protocol is applied are listed in Section 4.

3.1. Where each data source may enter

Table 2 lists the stages of the pipeline and the data each stage may read. "Log" is the biased training log, which holds the exposure matrix and the outcomes of the exposed pairs. "Validation" and "test" are the parts of the uniformly collected data that a split assigns to each role.

The additive imputation is fitted on the log without cross-fitting, so the same log supplies $\hat{Y}$ and the training data. The DRGP prior of Section 4.9 is cross-fitted over ten folds before its coefficients are fitted. Test scores are computed after every epoch and stored, but no choice reads them. The sensitivity analysis of the results reads them only to measure what selection on test data would change, and it does not report such a selection as a method.

The Coat exception.. The shipped Coat propensities were estimated by the dataset authors from a uniformly collected sample (Schnabel et al., 2016). That sample is also the source of our validation and test data, so uniform data reach a training input on Coat, indirectly. Every propensity-based row on Coat uses these propensities: DR-LightGCN, NAVIP, DR-JL, MRDR, DR adjacency, DRUP and the prior rows. To check the effect on the prior rows, we refitted them with propensities estimated from the log by the logistic model of Section 4.4. Table 9 gives the results. Relative to their no-prior backbones, the prior gains are $+19.44 \times 10^{-3}$ [18.66] for iALS-p and $+33.62 \times 10^{-3}$ [16.72] for LightGCN-p with estimated propensities. With the shipped propensities they are $+16.82 \times 10^{-3}$ [18.04] and $+31.98 \times 10^{-3}$ [20.65]. The paired differences between the two propensity inputs are $+2.63 \times 10^{-3}$ [5.27]

for iALS-p and $+1.64 \times 10^{-3}$ [10.30] for LightGCN-p, with $p_{\text{Holm}} = 1.000$ in both cases. The values in brackets are 95% half-widths. The stored no-prior runs of iALS and LightGCN serve as the reference for both settings, because neither reads the propensities in its loss. The check changes only the propensity input of the prior rows. For LightGCN-p the gain with estimated propensities is not removed. For iALS-p the estimated-propensity gain is not significant ($p_{\text{Holm}} = 0.33$), so the statement that the gain remains is a point estimate only. The check does not test the assumption that exposure is independent of the outcome given the covariates, which the closed-form posterior of the method requires. It also does not cover the other propensity-based rows of Coat: DR-LightGCN, NAVIP, DR-JL, MRDR, DR, DRUP and the QP-LightGCN family were not refitted, so their dependence on the shipped propensities is not checked.

3.2. Splits

On Coat, the split is made within each user: 30% of each user's candidate items go to validation and the rest go to test. On Yahoo! R3, KuaiRand-Pure and KuaiRec, the split is made over users: 30% of the test users go to validation, and no user is in both parts of a split. Each split is a function of its seed. Coat uses ten splits (seeds 0 to 9) and the other datasets use five (seeds 0 to 4). On Yahoo! R3, a fixed random 5% of the users (270) is excluded from validation and testing.

A user is a test user of a split only if the user has a test entry in that split. A user can fall in the validation part of every split, so the number of test users in the files of the final method is 225, 2,310, 8,321 and 1,410 on Coat, Yahoo! R3, KuaiRand-Pure and KuaiRec. These are slightly below the evaluated-user counts of Table 3, and equal to them on Coat.

Candidates are ranked for each user within the part being scored, and users with no positive candidate in that part are skipped. We rank all candidates and do not sample negatives, because sampled metrics do not agree with the full ranking metric (Krichene and Rendle, 2020). The metric is nDCG@5 on Coat and Yahoo! R3, nDCG@10 on KuaiRand and nDCG@20 on KuaiRec. Tables report the mean over splits and the standard deviation over splits with ddof = 0. The ensemble stage of Section 4.8 asserts, for every split it reads, that validation and test users are disjoint (on Yahoo! R3, KuaiRand and KuaiRec) or that validation and test items are disjoint for each user (on Coat).

3.3. Training and per-split selection

Each configuration is trained once on the log, and the same trained model is scored on every split. The splits therefore share training and differ only in selection. After each epoch, the model is scored on the validation and test parts of every split. Each split keeps the (configuration, epoch) pair with the highest validation nDCG, and the first maximum wins a tie. A split stops when its validation score has not improved for the patience period: five epochs on Coat and Yahoo! R3, and three epochs on KuaiRand and KuaiRec. The shared run ends when every split has stopped or the epoch limit is reached. Once a split has stopped, its selection is fixed. Later epochs are stored in the banks but cannot be chosen.

The gradient-trained models use epoch limits of 100 (Coat), 60 (Yahoo! R3), 40 (KuaiRand) and 30 (KuaiRec). Batch sizes are 256, 1,024, 8,192 and 32,768. Within a dataset, all methods share the epoch limit, batch size and patience, except AdvInfoNCE on KuaiRec, which uses batch size 4,096. On KuaiRec, each training epoch is capped at 2,000,000 pairs.

Three further quantities are computed from the same banks for the sensitivity analysis. Each is defined on the rows that the stopping rule of its split allows to be chosen.

- The per-split test oracle is the largest test value of a split over all configurations and epochs. It chooses on test data.
- The configuration-level oracle is the configuration whose mean test value, at its own validation-selected epoch, is largest. It also chooses on test data.
- The expected maximum of the validation-selected test value over k configurations is computed as a function of k (Dodge et al., 2019), and the share of 1,000 random subgrids of size $k_{\min}$ in which the sign of a primary comparison flips is reported. Here $k_{\min}$ is the smaller of the two grids of the comparison.

The first two are diagnostics. They are never reported as the result of a method.

3.4. Reconciliation of the stored results

Before the analysis, the selection of each (file, method) pair was replayed from its stored bank. The replay applies the training-time rule of Section 3.3,

including the patience value recorded in the bank, and its mean test score is compared with the stored mean. Of the 154 (file, method) pairs in the reconciliation, 151 agree within 0.002 in nDCG. The three that do not agree are DR-LightGCN on KuaiRec (replayed mean higher by 0.0033), DR-LightGCN on Yahoo! R3 (higher by 0.0058) and MACR on Yahoo! R3 (higher by 0.0060). All three come from two unsplit legacy result files that do not record the patience value, and the stored selections are reproduced only under patience values of 3 (KuaiRec) and 5 (Yahoo! R3). These three stored values are used in the tables: the KuaiRec DR-LightGCN value enters the P1 comparison of Table 6, and the Yahoo! R3 DR-LightGCN and MACR values appear in Table 4. Their selections cannot be reproduced from the banks, so we report the stored values. Every sensitivity number in the results is reported relative to the reconciled selection.

3.5. Seeds

Each split's selected configuration is retrained with seeds 1, 2 and 3. Each split re-selects its epoch on its own validation part, and the configuration is not changed. The retrained final method is compared with the two baselines that had the highest main-run test means in each dataset: iALS and DR-LightGCN on Coat, r-AdjNorm and LightGCN on Yahoo! R3 and KuaiRand, and DR-JL and MRDR on KuaiRec. The baselines are retrained in the same way. Because these two baselines were chosen on test data, the seed comparisons are descriptive, and they are reported without a significance claim beyond the Holm-adjusted values given in the text; the Holm correction runs over the two tests of each dataset. DRUP is deterministic and is not retrained. The seed-0 values in the tables are the main-run values.

Seeds 1 to 3 were not run for the Coat prior rows (iALS-p and LightGCN-p). Their Coat values come from seed 0 only. The result files also contain re-runs of split 0 with seeds 100 to 102, which are stored separately. These runs are not used in any table.

3.6. Statistical tests

Differences are computed per user. For each user, the test score is averaged over the splits in which the user is a test user. For a comparison $A - B$, let d_u be the per-user difference for user u. The mean $\bar{d}$ over users is the reported difference, and a paired t test over users gives the two-sided p-value and a 95% half-width. Differences are reported as $\times 10^{-3}$ with the half-width in brackets. The reported difference is not the difference of the two table

means. The two can have opposite signs. For example, on Coat the difference of the table means of DR-LightGCN and LightGCN is $+4.1\times10^{-3}$, while the mean per-user difference is -0.65×10^{-3} [30.16]. The per-user difference is the statistic used throughout.

A difference is called significant only when its Holm-adjusted p is below 0.05. The families are as follows.

Primary family, per dataset. Four comparisons, P1 to P4, are defined in code from the stored results. P1 and P4 choose their third-party debiaser and trained comparator, respectively, by mean validation score, and never by test score.

- P1: the third-party debiaser with the highest mean validation score on the dataset, against LightGCN.
- P2: each simple prior against its backbone, that is iALS-p against iALS and LightGCN-p against LightGCN, where both exist on the dataset.
- P3: the best refinement against the simple prior of the same learner. On Coat this gives two comparisons, the QP-LightGCN family against LightGCN-p and DR-iALS against iALS-p. On KuaiRec it gives the QP-LightGCN family against LightGCN-p.
- P4: DRUP against the trained model with the highest mean validation score. This is computed on KuaiRec only.

The primary families contain five comparisons on Coat and KuaiRec and two on Yahoo! R3 and KuaiRand.

Walk-correction family. DRUP against DR adjacency, one comparison per dataset.

Refinement families. Up to three families per dataset. The first holds each refinement against the simple prior of the same learner. The second, on Coat only, holds DR-iALS against iALS-p. The third holds the shuffled and popularity controls against LightGCN, on the datasets where those controls were run.

Backbone family. Every method against LightGCN, reported in Table A.12 to Table A.15 in the appendix.

The Nadeau and Bengio correction (Nadeau and Bengio, 2003) is applied as a second test to the split means, and we describe the variant we use. With J splits, split-level differences e_j and $\rho = 0.3$ the validation fraction, the variance is $\hat{\sigma}^2 = s_e^2\,(1/J + (1-\rho)/\rho)$, where s_e^2 is the sample variance of the e_j. The statistic $\bar{e}/\hat{\sigma}$ is compared with a t distribution with $J-1$ degrees of freedom, and Holm adjustment is applied to these second p-values within the same families. This is a heuristic correction, and it is not the standard form. The standard form uses the ratio of test to training size, n_2/n_1, and here the ratio is $(1-\rho)/\rho = n_1/n_2$. The models are trained once and only the selection and test splits vary, so the standard derivation does not apply directly. With $J = 5$ the variance factor is $1/5 + 7/3 \approx 2.53$, against $1/5 + 3/7 \approx 0.63$ under the standard ratio, so the correction is conservative for these splits. We report its values beside the standard paired test and do not treat either as exact.

Equivalence is assessed with two one-sided tests (Schuirmann, 1987) at margin δ, each at level 0.05. The two null hypotheses are $\bar{d} \le -\delta$ and $\bar{d} \ge \delta$. The reported TOST p-value is the larger of the two one-sided p-values. The margins are ± 0.010 nDCG@5 on Coat and Yahoo! R3, and ± 0.003 nDCG on KuaiRand (nDCG@10) and KuaiRec (nDCG@20). They are sensitivity thresholds, chosen before the analysis and not derived from the literature, and TOST is presented as a sensitivity analysis of these thresholds. On Yahoo! R3, KuaiRand and KuaiRec, a TOST result below 0.05 is stated as equivalence within the margin. On Coat, the same result is stated only as the exclusion of differences larger than the margin in absolute value, and equivalence is not claimed. A result inside a margin is never stated as equality.

The minimum detectable effect at 80% power and two-sided level 0.05 is $\mathrm{MDE} = (t_{0.975,\,n-1} + t_{0.80,\,n-1})\,s_d/\sqrt{n}$, where n is the number of users, s_d is the standard deviation of the per-user differences, and $t_{q,\nu}$ is the q quantile of a t distribution with ν degrees of freedom. The MDE uses only the user-level standard deviation and ignores the dependence between splits, so it is optimistic. For the primary comparisons in the analysis, the MDE is 11.8 to 43.1×10^{-3} on Coat, 4.7 to 5.6×10^{-3} on Yahoo! R3, 1.5 to 1.9×10^{-3} on KuaiRand and 0.7 to 4.9×10^{-3} on KuaiRec. The Coat values are large because the Coat analysis has 225 evaluated users, and an MDE of this size bounds what the Coat data can show.

Each user contributes one difference per comparison, which is the mean over the splits in which the user is a test user. The splits share users and candidates, so the paired t test over overlapping splits can be anti-

conservative (Nadeau and Bengio, 2003). For this reason the Nadeau and Bengio column is reported beside it for the primary comparisons. The tests treat the training log, the fitted nuisances and the trained models as fixed.

4. Methods evaluated

Every method is run by us on the splits and the data preparation of Section 3. No number in this study is taken from the literature. Configuration counts are the numbers stored in the result files. Each count is the product of the grid values listed below, and those values are the arguments of the run scripts of the repository. Table 4 lists the counts of every method in the main comparison.

Four refinements of the additive prior (Sections 4.6 to 4.9) were developed by us in earlier versions of this work, as candidate methods. Here they are reported as one family, with every configuration count, and the members that serve as controls are reported as controls. None of them is claimed as a contribution.

4.1. Datasets

We use four public benchmarks. Each pairs a biased log with a test set collected without the selection of the log, so that a debiasing method can be measured against unbiased outcomes. Table 3 gives the sizes, which were checked against the data loaders of the repository.

Coat (Schnabel et al., 2016) has 290 users and 300 items. The log holds 6,960 self-selected ratings (8.0% of the matrix), of which 1,905 are ratings of 4 or more. The test set holds 16 ratings per user on uniformly random items. Candidates are the items a user did not rate in the log, 14.7 per user on average. The dataset ships propensities, which its authors estimated from a uniform sample; Section 3 describes how they are used.

Yahoo! R3 (Marlin and Zemel, 2009) has 15,400 users and 1,000 songs. The log holds 311,704 self-selected ratings (2.0% dense). The test set holds 54,000 ratings of ten random songs for 5,400 users, in the version distributed with (Chen et al., 2021). A fixed random 5% of these users (270) is excluded from validation and testing. The remaining 5,130 test users have ten candidates each, and 2,315 of them have a positive candidate.

KuaiRand-Pure (Gao et al., 2022b) is the only benchmark whose log records platform exposure. An impression is an item shown in the standard feed, and the outcome is a click. We keep 10,000 users who also received

randomly exposed videos, and all 7,583 videos. The log holds 511,863 impressions (0.68% dense), 47% of them clicks. The test set consists of the randomly exposed impressions of the same period, restricted to videos that the user was not shown in the standard log. Users have 43.6 candidates on average, and 8,347 of them have a positive candidate.

KuaiRec (Gao et al., 2022a) provides a dense log of 10.3 million views by 7,176 users of 10,728 items (13.4% dense). Its test set is a fully observed matrix for 1,411 users and 3,327 items. Each user is ranked over the test items that the user never viewed in the log, 3,314 on average.

Positive outcomes are ratings of 4 or more (Coat, Yahoo! R3), a click (KuaiRand) and a mean watch ratio of at least 2 (KuaiRec). On Coat and Yahoo! R3, "exposure" means that the user chose to rate the item, so the propensity models self-selection and not the allocation of a platform. The candidate sets of these two datasets hold about ten to fifteen items, which limits the range that nDCG can span on them.

4.2. Training-free operators and trained baselines

The training-free operators are computed from the log and need no training. Pop ranks items by their popularity in the log. Impute ranks by the additive imputation of Section 4.4 alone. Obs, EASE, GF-CF, IPS, DR, DRUP, EASE-DR and GF-CF-DR are further training-free baselines, with the configuration counts of Table 4. DR and DRUP are defined here because the ensemble of Section 4.8 uses DRUP. The doubly robust edge estimate is

$$W_{ui} = \hat{Y}_{ui} + \frac{O_{ui}(Y_{ui} - \hat{Y}_{ui})}{\bar{p}_{ui}}, \tag{2}$$

which is unbiased for Y_{ui} under correct propensities. Let $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$, where d_u and d_i are the row and column sums of W or of $\hat{Y}$ (a grid choice), and let α be a grid value. DRUP scores a pair by its one-hop term and its three-hop term, which is a walk-corrected propagation on the weighted graph $C \circ W$:

$$S_{ui} = \frac{(C \circ W)_{ui}}{\overline{|C \circ W|}_u} + \beta \frac{T_{ui}}{\overline{|T|}_u}, \tag{3}$$

where T is the three-hop term with the walks that reuse an edge replaced by their unbiased counterparts, and $\overline{|\cdot|}_u$ is the mean absolute value over the items of user u. DR adjacency is the same operator without the walk correction. The weight β and the other grid values are chosen on validation

data. The training-free grids are the products of the following values, as run by the repository. On Coat, the propensity floor takes five values, the imputation penalty three, the imputation source two, α three, the degree source two and β eight values, which gives $5 \times 3 \times 2 \times 3 \times 2 \times 8 = 1{,}440$ for DR and for DRUP. On Yahoo! R3, the floor takes two values, the imputation penalty one, α two, the control-variate weight of the DR estimate four, the imputation source two, the degree source two and β eight values, which gives $2 \times 1 \times 2 \times 4 \times 2 \times 2 \times 8 = 512$. On KuaiRec, the grid has three floors, two values for each of the other terms except β, and the eight values of β, which gives 384.

The trained baselines are the following. Their grids are given per dataset, and the learning rate and weight decay are shared by all trained models unless stated otherwise.

Coat.. The grid is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}, 10^{-3}\}$ and width in $\{32, 64\}$. For LightGCN, LightGCN-pt, r-AdjNorm, NAVIP, DR-LightGCN and SimGCL the depth ranges over $\{1, 2, 3\}$. Epochs are limited to 100, the batch size is 256 and the patience is 5. Configurations: LightGCN 54, LightGCN-pt 54, NAVIP 54, DR-LightGCN 54, r-AdjNorm 108 (with $r \in \{0.3, 0.7\}$), SimGCL 108 (with weight $\lambda_{\text{cl}} \in \{0.1, 0.5\}$), MACR 36 (with $c \in \{0.1, 0.3\}$), PDA 36 (with $\gamma \in \{0.1, 0.2\}$), MF, IPS-MF, DR-MF, BPR-MF, DR-JL and MRDR 18 each, iALS 24 (with $\alpha \in \{3, 10, 30\}$, regularisation in $\{1, 10, 100, 1000\}$ and width in $\{32, 64\}$) and iALS grid B 18 (with $\alpha \in \{10, 30, 100\}$ and regularisation in $\{10, 100, 300\}$). The iALS runs of the prior family use an epoch limit of 30 on Coat, and the stored iALS run uses 100. The epoch limits are therefore not the same across the iALS rows.

Yahoo! R3.. The grid is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}\}$ (with 10^{-3} added for the matrix models), width 64 and depth 2. For LightGCN, NAVIP and r-AdjNorm the width ranges over $\{64, 128\}$ and the depth over $\{1, 2, 3\}$. Epochs are limited to 60, the batch size is 1,024 and the patience is 5. Configurations: LightGCN 36, NAVIP 36, r-AdjNorm 72, LightGCN-pt 6, DR-LightGCN 6, SimGCL 12, MACR 18, PDA 18, iALS 12 (with the iALS grid above and width 64), MF, IPS-MF, DR-MF, BPR-MF, DR-JL and MRDR 9 each.

KuaiRand-Pure.. The grid is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$ and weight decay in $\{10^{-5}, 10^{-4}\}$, with width 64 and depth 2 for every model. Epochs

are limited to 40, the batch size is 8,192 and the patience is 3. Configurations: LightGCN 6, LightGCN-pt 6, NAVIP 6, DR-JL 6, MF 6, DR-MF 6, BPR-MF 6, r-AdjNorm 12, SimGCL 12, MACR 12 and iALS 12. MRDR, IPS-MF, PDA and DR-LightGCN were not run on this dataset.

KuaiRec.. The grid is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$ and weight decay in $\{10^{-5}, 10^{-4}\}$. For LightGCN and LightGCN-pt the width ranges over $\{64, 128\}$ and the depth over $\{1, 2, 3\}$, and the other models use width 64 and depth 2. Epochs are limited to 30, the batch size is 32,768, the patience is 3, and each epoch is capped at 2,000,000 pairs. Configurations: LightGCN 36, LightGCN-pt 36, r-AdjNorm 12, NAVIP 6, DR-LightGCN 6, SimGCL 12, MACR 12, iALS 12, MF 6, IPS-MF 6, DR-MF 6, BPR-MF 6, DR-JL 6 and MRDR 6. PDA (12 configurations) and two pointwise negative-sampling variants, LightGCN-pn and DR-LightGCN-pn (24 configurations each), were also run on KuaiRec and are not in the main comparison.

Across datasets, the graph models were given depth grids $\{1, 2, 3\}$ on Coat, Yahoo! R3 and KuaiRec, and only depth 2 on KuaiRand. Budgets are therefore unequal, and the counts make the difference visible. The selection rule of Section 3.3 is applied to each grid as it stands, and the sensitivity of the conclusions to the size of the grid is measured in the results.

4.3. AdvInfoNCE re-implementation

AdvInfoNCE (Zhang et al., 2023) is an adversarial contrastive loss. We re-implemented it on the LightGCN encoder. Our numbers are for this re-implementation and not for the original method, and the official code was not run. The main simplifications are these.

- The score is the cosine similarity of the propagated embeddings divided by a temperature τ. Each positive is contrasted with 64 sampled negatives.

- The adversary is a pair of embedding tables (users and items, dimension 64) trained with Adam to maximise the contrastive loss. Its learning rate is 10^{-2} on Coat, 10^{-3} on Yahoo! R3 and KuaiRand, and $5 \cdot 10^{-4}$ on KuaiRec. Only this embedding-based adversary is implemented.

- The adversary budget of the original formulation is replaced by a number of adversarial rounds $\eta \in \{4, 12\}$. A round runs after each epoch from the second epoch on and lasts at least one pass over the positives

and 100 Adam steps. The adversary's KL divergence is logged but not projected onto a ball.

The grids differ in size, and the budgets are unequal. Coat has 216 configurations: three learning rates, three weight decays, widths $\{32, 64\}$, depths $\{1, 2, 3\}$, $\tau \in \{0.2, 0.75\}$ and $\eta \in \{4, 12\}$. Yahoo! R3 has 16 configurations with $\tau \in \{0.1, 0.28\}$, $\eta \in \{4, 12\}$, learning rates $\{3 \cdot 10^{-3}, 10^{-3}\}$, weight decays $\{10^{-5}, 10^{-4}\}$, width 128 and depth 2. KuaiRand has 8 configurations with $\tau \in \{0.1, 0.28\}$, learning rates $\{10^{-2}, 3 \cdot 10^{-3}\}$, weight decay 10^{-4}, width 64 and depth 2. KuaiRec has 8 configurations with $\tau \in \{0.2, 2.0\}$, learning rates $\{10^{-2}, 3 \cdot 10^{-3}\}$, weight decay 10^{-5}, width 64 and depth 2. KuaiRand and KuaiRec both use $\eta \in \{4, 12\}$. On Coat, AdvInfoNCE thus has 216 configurations against 54 for LightGCN and 108 for r-AdjNorm. A larger grid gives a method more chances to fit noise in the validation data, so the Coat grid favours AdvInfoNCE and the smaller grids work against it. We did not match budgets for this baseline. The only wider re-tuning in this study is the Coat iALS grid (Table 10).

4.4. The additive prior and the learners iALS-p and LightGCN-p

Exposure propensity.. On Yahoo! R3, KuaiRand-Pure and KuaiRec the exposure propensity is a logistic model fitted to the exposure matrix of the log,

$$\hat{p}_{ui} = \sigma\big(a + b\,\tilde{\ell}_u + c\,\tilde{\ell}_i\big),$$

where $\tilde{\ell}_u$ and $\tilde{\ell}_i$ are standardised logits of the user and item exposure rates. The coefficients are fitted by Newton's method on all mn pairs. On Coat, the shipped propensities are used instead. Every propensity is then truncated below at a floor f, which gives $\bar{p}_{ui} = \max\{\hat{p}_{ui}, f\}$. The floor is an argument of each run and is not part of any configuration grid. Its values are 0.05 for every LightGCN-p and QP-LightGCN run on all four datasets. The exceptions are the following. On Coat, iALS-p and DRGP-iALS use 0.1, the prior alone uses 0.1, and the DRGP-LGN learners use 0.1. On Yahoo! R3, the prior alone uses 0.1. On KuaiRand, the prior alone uses 0.02. On KuaiRec, iALS-p, the prior alone and the DRGP-LGN learners use 0.2. The DRGP prior alone was run at floors 0.05, 0.1 and 0.2 on Coat and at 0.1 and 0.2 on KuaiRec. The result files do not record a rule for these values, The floors of the training-free operators are part of their grids: Coat 0.01 to 0.2, Yahoo! R3 0.1 and 0.2, and KuaiRec 0.05 to 0.2.

Additive imputation.. The imputed outcome is $\hat{Y}_{ui} = \min\{1, \max\{0, \mu + b_u + b_i\}\}$. Its parameters are fitted to the logged outcomes $O \odot Y$, with weights $O/\bar{p}$, by 20 alternating closed-form ridge sweeps with penalty $\lambda = 5$. The fit uses the log only and is not cross-fitted, so the same log supplies $\hat{Y}$ and the training data.

Prior.. Let $\ell_{ui} = \operatorname{logit} \hat{Y}^c_{ui}$, where $\hat{Y}^c$ is $\hat{Y}$ clipped to $[10^{-3}, 1 - 10^{-3}]$. The prior is the standardised logit $z_{ui} = (\ell_{ui} - \bar{\ell})/s_\ell$, where the mean and the standard deviation are taken over all mn pairs. Within a user, the ranking by z is the ranking by b_i, up to ties created by the clipping of $\hat{Y}$ to $[0, 1]$. The prior alone is a training-free scorer (one configuration), and it is reported in Table 7.

iALS-p.. The iALS objective of Hu et al. (Hu et al., 2008) receives a fixed offset,

$$\sum_{u,i} c_{ui} \big(y_{ui} - \beta z_{ui} - x_u^\top y_i\big)^2 + \operatorname{reg}\left(\|x_u\|^2 + \|y_i\|^2\right),$$

where $c_{ui} = 1 + \alpha$ on logged positives and $c_{ui} = 1$ elsewhere. The grid is $\alpha \in \{10, 30, 100\}$, regularisation in $\{10, 100, 300\}$, $\beta \in \{0.1, 0.2, 0.3, 0.5, 1\}$ and width in $\{32, 64\}$, which gives 90 configurations on Coat. On KuaiRec the main-table row uses an eight-configuration grid, $\alpha \in \{10, 30\}$, regularisation in $\{100, 1000\}$, $\beta \in \{0.5, 1\}$ and width 64. The grid of the DRGP queue has 45 configurations on KuaiRec ($\alpha \in \{10, 30, 100\}$, regularisation in $\{10, 100, 300\}$, the five values of β, width 64). It is reported as a reference row.

LightGCN-p.. The offset enters the BPR margin of the LightGCN learner, so that $s_{ui} - s_{uj} = f_\theta(u, i) - f_\theta(u, j) + \beta(z_{ui} - z_{uj})$, with $\beta \in \{0.5, 1\}$. The grids are as follows. On Coat, learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}, 10^{-3}\}$, width in $\{32, 64\}$ and depth in $\{1, 2, 3\}$, which gives 108 configurations. On Yahoo! R3, learning rate $3 \cdot 10^{-3}$, weight decay in $\{10^{-5}, 10^{-4}\}$, width in $\{64, 128\}$ and depth in $\{1, 2, 3\}$, which gives 24. On KuaiRand, learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}\}$, width 64 and depth 2, which gives 12. On KuaiRec, learning rate $3 \cdot 10^{-3}$, weight decay in $\{10^{-5}, 10^{-4}\}$, width in $\{64, 128\}$ and depth in $\{1, 2, 3\}$, which gives 24.

4.5. Prior variants

The variants below change the additive imputation or the way it enters the score. Each is defined in the code of the repository. In every variant, the prior is the standardised logit of the variant's imputation, except for `pop`, which replaces the offset by a popularity score.

add. The imputation of Section 4.4, fitted with weights $O/\bar{p}$.

add_noips. The same ridge fit with weights O in place of $O/\bar{p}$, so that the fit uses the exposed pairs without inverse-propensity weights.

pop. The imputation, and therefore Q, is unchanged. The offset is replaced by the standardised logarithm of the number of positives of each item in the log, $\log(1 + \sum_v O_{vi} Y_{vi})$, broadcast to every user.

shuf_item. The item biases b_i are permuted across items by a fixed permutation (seed 0). The parameters μ and b_u are kept, and the imputation and the prior are recomputed from the permuted biases.

user_only. The item biases are set to zero, so every item has the same score within a user. The ranking within a user is then decided by tie-breaking.

In the learners of this analysis, only the prior enters the score. The variant's Q is used only by the Q-weighted methods, and the prior-anatomy runs do not use them.

The prior-anatomy runs are the following. On Coat, iALS-p and LightGCN-p are run over the ten splits with the variants add, add_noips, pop and shuf_item. On KuaiRec, iALS-p is run over the five splits with the same four variants on the eight-configuration grid. The prior alone is run for all five variants on all four datasets, with the floors listed in Section 4.4. The floor of the Coat LightGCN-p runs is 0.05, and that of the Coat iALS-p runs is 0.1.

4.6. Q-weighted negatives

The probability that an unlogged pair is a positive is

$$Q_{ui} = \frac{\hat{Y}_{ui}(1 - \bar{p}_{ui})}{1 - \bar{p}_{ui}\hat{Y}_{ui}}, \tag{4}$$

clipped to $[0, 1]$. This is the posterior under the assumption that exposure and outcome are independent given the covariates on which the propensity depends.

The Q-weighted BPR loss uses only logged positives. For each positive (u, i) one negative j is drawn uniformly from the items, and it is redrawn once if (u, j) is itself logged. The negative is weighted by $w_{uj} = \max\{1 - f_n Q_{uj}, 10^{-3}\}$, and the loss over a batch B is

$$\mathcal{L} = -\frac{\sum_{(u,i,j)\in B} w_{uj} \log \sigma\big(s_{ui} - s_{uj}\big)}{\sum_{(u,i,j)\in B} w_{uj}} + \lambda \frac{1}{|B|} \sum_{(u,i,j)\in B} \big(\|e_u\|^2 + \|e_i\|^2 + \|e_j\|^2\big),$$

where e_u, e_i and e_j are the embeddings and λ is the weight decay. The normalisation by the total weight keeps the loss scale independent of how many negatives are down-weighted. The factor f_n is 1 in every job except the Coat job without offset, where $f_n \in \{0.5, 1\}$. The encoder is LightGCN.

The Q-weighted iALS (DR-iALS) uses the same weights as confidences. Logged positives get $c_{ui} = 1 + \alpha$, and unlogged pairs get $c_{ui} = \max\{1 - f_n Q_{ui}, 10^{-3}\}$. Its grid is $\alpha \in \{3, 10, 30\}$, regularisation in $\{1, 10, 100, 1000\}$, $f_n \in \{0.5, 1\}$ and width in $\{32, 64\}$, which gives 48 configurations on Coat.

The configuration counts of the Q-weighted LightGCN (DR-BPR-LGN) are 108 on Coat (with $f_n \in \{0.5, 1\}$), and DR-BPR-LGN-1 (with $f_n = 1$) has 54 on Coat, 36 on Yahoo! R3, 6 on KuaiRand and 24 on KuaiRec.

A low-rank variant replaces $\hat{Y}$ in Q, and in the prior of its offset variant, by a rank-16 low-rank imputation with ridge penalty 5. It has 54 configurations on Coat in the Q-only form and 108 in the offset form.

Three control weightings replace Q in the same learner. The first is Q computed from $\hat{Y}$ alone, with no propensity correction (DR-BPR-yhat; 54 configurations on Coat). The second is Q computed from item popularity, rescaled to the mean of the DR weight (DR-BPR-pop; 54 on Coat and 12 on Yahoo! R3). The third is a fixed random permutation of the values of Q over the matrix (DR-BPR-shuf; 54 on Coat and 12 on Yahoo! R3). The controls test whether the information in Q matters, and they are not proposed methods.

4.7. QP-LightGCN

QP-LightGCN combines the Q weighting with the offset of Section 4.4, so that $f_n = 1$ and the BPR margin contains $\beta(z_{ui} - z_{uj})$. Two offset grids are used. DR-BPR-LGN-p has $\beta \in \{0.5, 1\}$, and DR-BPR-LGN-p2 has

$\beta \in \{1, 2\}$. The configurations of each offset grid are 108 on Coat, 24 on Yahoo! R3 (learning rate $3 \cdot 10^{-3}$) and 12 on KuaiRand. On KuaiRec, DR-BPR-LGN-p has 24 configurations (learning rate $3 \cdot 10^{-3}$) and DR-BPR-LGN-p2 has 72 (learning rates 10^{-2}, $3 \cdot 10^{-3}$ and 10^{-3}).

The QP-LightGCN family of each dataset pools its jobs. For each split, the job, configuration and epoch with the highest validation score are selected. The family contains DR-BPR-LGN, DR-BPR-LGN-p and DR-BPR-LGN-p2 on Coat, which gives 324 configurations. On Yahoo! R3 it contains DR-BPR-LGN-1, DR-BPR-LGN-p and DR-BPR-LGN-p2, which gives 84 configurations. On KuaiRand and KuaiRec the family contains the same job types, DR-BPR-LGN-1, DR-BPR-LGN-p and DR-BPR-LGN-p2, and it has 30 and 120 configurations respectively.

4.8. Ensemble with DRUP

For each split, component A is the QP-LightGCN family configuration selected for that split, and component B is the DRUP configuration selected for that split. The scores of each component are standardised within each user over the unlogged items of the training log, that is, over the pairs with $O_{ui} = 0$. The standardised score is $\tilde{S}_{ui} = (S_{ui} - \bar{S}_u)/s_u$, with the mean and the standard deviation taken over those pairs. The ensemble score is

$$(1 - w)\, \tilde{S}^A_{ui} + w\, \tilde{S}^B_{ui}, \qquad w \in \{0, 0.1, \ldots, 1\}.$$

The weight w is the first maximiser over this grid of the validation nDCG, and the test score is computed with that w. The ensemble stage also stores the rows $w = 0$ (component A), $w = 1$ (DRUP) and $w = 0.5$ as checks. The weight is chosen from validation lists only. The ensemble adds no configuration of its own, so the components carry the configuration counts.

4.9. DRGP: a leave-edge-out DR graph prior

DRGP has two stages. It was run on Coat and KuaiRec only, and its runs have not been replicated over seeds. On KuaiRec, 5 of 5 jobs completed, with 57 configurations in total, in a single run without seed repeats.

Stage A: the prior.. For each pair, define the leave-edge-out 3-hop term of the DR graph. Let C be the degree weights defined above, with the degrees taken from $\hat{Y}$. Let $\mathrm{sq} = C \circ C \circ W$ and $A_{ui} = \sum_j \mathrm{sq}_{uj} + \sum_v \mathrm{sq}_{vi} - 2\,\mathrm{sq}_{ui}$. Then

$$T^{\mathrm{LEO}}_{ui} = T_{ui} - (W_{ui} - \hat{Y}_{ui})\,(C_{ui} A_{ui} + C^3_{ui}),$$

where T is the walk-corrected three-hop term on $C \circ W$. The term T^{LEO}_{ui} does not depend on the exposure and outcome of the pair (u, i). The same construction applied to the observed positive graph $O \circ Y$, with base 0, gives a third feature.

The features are $\phi_1 = \operatorname{logit} \hat{Y}$, ϕ_2 and ϕ_3 equal to T^{LEO} of the DR graph with $\alpha \in \{0.5, 0.7\}$, and ϕ_4 equal to T^{LEO} of the observed positive graph with $\alpha = 0.5$. Each feature is standardised over the matrix. The prior is

$$\pi_{ui} = a_u + \sum_k \theta_{k,b(u)}\, \phi_{k,ui},$$

where $b(u)$ is a user-activity bucket defined by the quantiles of each user's number of exposures. The coefficients minimise $\sum_{u,i}(\pi_{ui} - W_{ui})^2 + \rho\|\theta\|^2$ with $\rho = 1$. This objective equals the DR squared risk up to a constant for each pair. By the Frisch-Waugh theorem, it is a ridge regression of the row-demeaned W on the row-demeaned features, within each bucket. The imputation inside the features uses penalty $\lambda = 5$ on Coat and $\lambda = 20$ elsewhere. The nuisances of the prior are cross-fitted over ten folds. The prior is the standardised logit $z(\pi)$ of π, and the number of buckets is 1 or 2 on Coat and 4 on KuaiRec.

Stage B: the learners.. DRGP-iALS is iALS-p with the prior $z(\pi)$ as its offset. Its grid is that of iALS-p, with 90 configurations on Coat and 45 on KuaiRec. DRGP-LGN-bpr is LightGCN with the BPR margin and the offset $\beta z(\pi)$, with $\beta \in \{0.1, 0.2, 0.5, 1\}$; it has 72 configurations on Coat and 8 on KuaiRec. DRGP-LGN-dr is a LightGCN learner with a self-imputing pointwise doubly robust loss. Writing $\tilde{\pi} = \min\{0.99, \max\{0.01, \pi\}\}$ and ℓ for the binary cross-entropy on logits,

$$\frac{1}{mn} \sum_{u,i} \Big[\ell(s_{ui}, \tilde{\pi}_{ui}) + \frac{O_{ui}}{\bar{p}_{ui}} \big(\ell(s_{ui}, Y_{ui}) - \ell(s_{ui}, \tilde{\pi}_{ui}) \big) \Big].$$

Its score offset is $\operatorname{logit} \tilde{\pi}$ with weight 1, so the imputed term vanishes at initialisation. It has 18 configurations on Coat and 2 on KuaiRec. DRGP-LGN-hyb adds this loss to the BPR loss with weight $\eta \in \{0.3, 1\}$, and it has 36 configurations on Coat.

5. Results

This section answers the four research questions in turn. All test numbers are nDCG on uniformly exposed test data. Uniform data are used for vali-

dation and testing. The one exception is Coat's shipped propensities, which the dataset authors estimated from the uniform sample; Section 5.3 reports a refit for the prior rows with the estimator used on the other datasets. Otherwise nuisances are fitted on the exposed log, and every configuration and epoch is selected on validation data within each split. Table 4 gives the benchmark, with the mean and standard deviation over splits and the number of configurations searched for each method. Paired differences are per-user averages of the difference between two methods, taken over the users who appear in both test sets. They are given in units of 10^{-3} nDCG, with the 95% confidence-interval half-width in brackets. A difference is called significant only when its Holm-adjusted p-value is below 0.05, and the Holm family is stated wherever a p-value is given. The analysis uses 225 users on Coat, 2,310 on Yahoo! R3, 8,321 on KuaiRand-Pure and 1,410 on KuaiRec. There are 10 splits on Coat and 5 on the other three datasets. The metric is nDCG@5 on Coat and Yahoo! R3, nDCG@10 on KuaiRand-Pure and nDCG@20 on KuaiRec.

Table 6 reports the primary comparisons. P1 compares the strongest third-party graph debiaser with LightGCN, and P2 compares a simple debiased prior with its backbone; both are reported on all four datasets. P3 compares the best refinement with the simple prior of the same learner, and P4 compares training-free doubly robust propagation with the best trained model on KuaiRec. P3 is reported on Coat and KuaiRec only, and P4 on KuaiRec only. The equivalence (TOST) margins are ± 0.010 on Coat and Yahoo! R3 and ± 0.003 on KuaiRand-Pure and KuaiRec. They are sensitivity thresholds chosen before the analysis and are not derived from the literature, and the TOST results are presented as a sensitivity analysis. On Coat, the TOST column states only which differences the data exclude.

The appendix table of every method against LightGCN (Tables A.12 to A.15) gives the backbone family used in several statements below.

5.1. Effects of evaluation choices (RQ1)

Selecting on the test split inflates the reported scores, and the inflation differs between methods. Across the trained methods in the sensitivity analysis, the per-split test oracle exceeds the validation-selected score by a mean of +0.0207 on Coat (range +0.0097 to +0.0302), +0.0044 on Yahoo! R3 (range +0.0015 to +0.0088), +0.0012 on KuaiRand-Pure (range +0.0005 to +0.0022) and +0.0009 on KuaiRec (range +0.0000 to +0.0028). On Coat the inflation is largest for DR-JL (+0.0302) and smallest for iALS (+0.0097),

while LightGCN inflates by +0.0241. This is enough to change an order. Under validation selection iALS scores 0.5812 and LightGCN 0.5683. Under the per-split test oracle, which chooses on test data and is a diagnostic only, the scores are 0.5909 for iALS and 0.5924 for LightGCN, so the order reverses. The configuration-level oracle, which picks one configuration by its mean test score, keeps the validation order, with 0.5843 for iALS and 0.5807 for LightGCN. The two kinds of test oracle therefore do not agree on Coat. The reversal is a property of the per-split test oracle only.

The searched grids differ in size, from 6 configurations for several KuaiRand-Pure and KuaiRec baselines to 324 for the QP family on Coat. To measure how much this matters, we drew 1,000 random subgrids of $k_{\min}$ configurations for each comparison and counted how often its sign changed (Table 5). Flips are common for comparisons whose effects are near zero. The QP family against LightGCN-p changes sign in 35.3% of the subgrids on Coat and 30.9% on KuaiRec. The largest instability is DRUP against DR-JL on KuaiRec. At $k_{\min} = 6$, 78.4% of the subgrids flip the sign, and the split-mean difference at matched budget is -14.90, while the full-grid per-user difference is +4.38 [2.21]. DRUP is training-free but searches 384 configurations on KuaiRec, against 6 for DR-JL, so its advantage over DR-JL depends on the budget granted to each method. The matched-budget values in Table 6 are split means and carry no test.

The weighting unit can also change a sign. For DR-LightGCN against LightGCN on Coat, the split-mean difference at matched budget ($k = 54$) is +4.11, while the per-user paired difference is -0.65 [30.16]. Every test in this paper uses the per-user unit. The split-mean values are reported only as matched-budget descriptions, and they are not tested.

Retraining the selected configurations with seeds 1 to 3 changes two main-run conclusions (the seeds column of Table 6). These comparisons are descriptive, because the two baselines of each dataset were chosen on test data. On Coat, QP-LightGCN against iALS is +20.02 [19.74] in the main run ($p_{\text{Holm}} = 0.0469$, in a family of eight baselines). Over seeds 1 to 3 the difference is +3.39 [20.70] ($p_{\text{Holm}} = 0.747$, in a family of two), which is not significant. On KuaiRec the main-run deficits of QP-LightGCN against DR-JL and MRDR, -4.56 [2.24] and -4.34 [2.16], shrink over seeds 1 to 3 to -1.03 [1.64] ($p_{\text{Holm}} = 0.2213$) and -1.56 [1.47] ($p_{\text{Holm}} = 0.0760$). On Yahoo! R3 and KuaiRand-Pure the QP-LightGCN family exceeds r-AdjNorm by +14.76 [5.23] and by +2.31 [1.23] over seeds 1 to 3; these seed comparisons are descriptive and carry no significance claim. The seed-to-seed standard

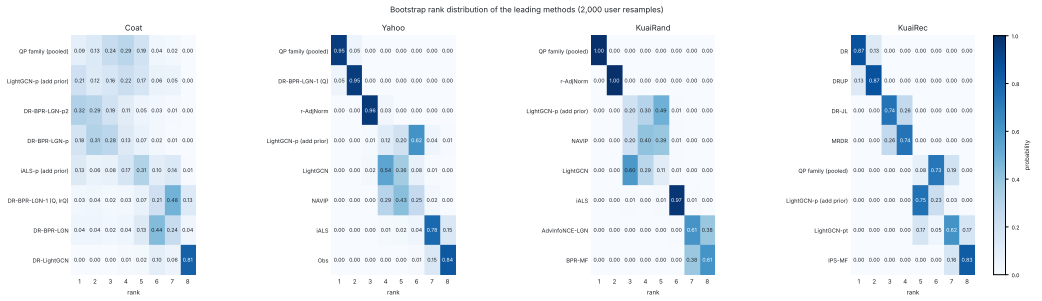


Figure 1: Bootstrap rank distributions over 2,000 user resamples. For each dataset the rows are the eight methods with the highest mean validation score, excluding control rows, and the columns are the rank by mean test score (1 is best). On Coat the splits are resampled together with the users. Each cell gives the probability of that rank.

deviation of QP-LightGCN on Coat (0.0049) is larger than that of iALS (0.0029), so the main-run Coat result is a single-run result.

Changing the metric changes 3 of the 14 pairwise conclusions in Table 5: the Coat comparisons of DR-LightGCN with LightGCN and of the QP family with LightGCN-p, and the KuaiRec comparison of DR-LightGCN with LightGCN. The alternative metric is nDCG@10 on Coat and Yahoo! R3, nDCG@20 on KuaiRand-Pure and nDCG@50 on KuaiRec. Of the 14 conclusions, 7 are significant at $p_{\text{Holm}} < 0.05$ in the main run. Their signs survive the per-split test oracle in 12 cases, the configuration-level oracle in 11, matched budget in 13 and the alternative metric in 11.

The bootstrap ranks (Figure 1) show how stable the leader is. On Yahoo! R3 the QP family ranks first in 95.0% of user resamples, and on KuaiRand-Pure in 99.8%. On KuaiRec, DR and DRUP have the same mean score (0.6378) and exchange first place, with probabilities 0.873 and 0.127. The bootstrap therefore cannot separate them. On Coat no method is a stable leader. The most frequent first place is DR-BPR-LGN-p2, with probability 0.319, and the QP family ranks first with probability 0.090.

5.2. What graph debiasing adds over an equally tuned backbone (RQ2)

On Coat the graph debiasers are not separated from LightGCN. The three with stored Coat results differ from it by +3.76 [5.74] (r-AdjNorm), −1.59 [10.13] (NAVIP) and −0.65 [30.16] (DR-LightGCN), and none of these is significant. For r-AdjNorm the TOST analysis at ±0.010 excludes differences

larger than the margin (TOST p = 0.017). For NAVIP (p = 0.0515) and DR-LightGCN (p = 0.271) it does not. The minimum detectable difference for the primary comparison P1 on Coat is 43.07 ($\times 10^{-3}$), so the study cannot exclude a gain of that size for DR-LightGCN. On Coat, iALS differs from LightGCN by +12.01 [16.12] (p = 0.143, not significant). The 95% half-widths of the three debiasers on Coat range from ±6 (r-AdjNorm) to ±30 (DR-LightGCN), so the Coat comparisons are underpowered.

On Yahoo! R3, r-AdjNorm is +4.10 [3.29] above LightGCN (p = 0.015, $p_{\text{Holm}} = 0.029$ in the two-comparison primary family). Adjusting over all comparisons with LightGCN (Table A.13) gives $p_{\text{Holm}} = 0.073$, and the Nadeau-Bengio correction for overlapping splits gives $p_{\text{Holm}} = 0.883$. On KuaiRand-Pure, r-AdjNorm is +1.99 [1.02] above LightGCN ($p_{\text{Holm}} < 0.001$). This difference is significant and also lies inside the sensitivity margin of ±0.003 (TOST p = 0.026), so the margin does not make it a null result. The Nadeau-Bengio correction gives $p_{\text{Holm}} = 1.000$. Across Coat, Yahoo! R3 and KuaiRand-Pure, then, the best third-party graph debiaser adds at most +4.10 ($\times 10^{-3}$) nDCG over the equally tuned backbone in the point estimate, and the Nadeau-Bengio correction removes its significance. The bound applies to third-party debiasers only. Our own QP-LightGCN family adds +16.77 [5.82] over LightGCN on Yahoo! R3 ($p_{\text{Holm}} < 0.001$; +20.02 over seeds 1 to 3) and +3.93 [1.53] on KuaiRand-Pure ($p_{\text{Holm}} < 0.001$; +4.87 over seeds 1 to 3). Under the Nadeau-Bengio correction neither difference is significant ($p_{\text{NB,Holm}} = 0.434$ and 1.000). These refinements are reported as members of the evaluated family in Section 5.3.

KuaiRec, whose test matrix is fully observed, differs. DR-LightGCN is +13.17 [3.31] above LightGCN ($p_{\text{Holm}} < 0.001$) and NAVIP is +10.41 [1.82] above it (Table A.15). Under the Nadeau-Bengio correction the DR-LightGCN difference has $p_{\text{NB,Holm}} = 0.787$. The pointwise LightGCN-pt, which has no debiasing, is +19.59 [4.55] above the BPR backbone. This is a difference of means: no paired test compares it with the +13.17 of DR-LightGCN, and the two intervals overlap. The training-free operators are stronger on this dataset (Section 5.4).

The re-implementation of AdvInfoNCE is below LightGCN on all four datasets: −46.47 [25.16] on Coat, −21.08 [7.78] on Yahoo! R3, −8.86 [2.50] on KuaiRand-Pure and −28.42 [6.46] on KuaiRec, each significant under the Holm adjustment of the backbone family (Table A.12 to Table A.15). SimGCL and MACR are below LightGCN on all four datasets and significantly so on Yahoo! R3, KuaiRand-Pure and KuaiRec. On Coat, their

differences (−17.49 [23.88] and −29.43 [24.60]) are not significant. These comparisons use our implementations, which are described in the protocol section.

Table 6 gives the primary comparisons with their Nadeau-Bengio columns. Under that correction, none of the four P1 comparisons is significant. The two P2 comparisons on KuaiRec remain significant, with $p_{\mathrm{NB,Holm}}$ of 0.009 for LightGCN-p and 0.043 for iALS-p. The P4 comparison, DRUP against DR-JL, is significant under the per-user test (+4.38 [2.21], $p_{\mathrm{Holm}} < 0.001$) but not under the correction ($p_{\mathrm{NB,Holm}} = 0.426$).

5.3. Which ingredient carries the gain (RQ3)

The simple prior is the additive imputation $\hat{\mu} + b_u + b_i$, fitted once on the exposed log with inverse-propensity weights and added to the score as an offset. Within a user its ranking is the ranking of the item term b_i. Used as an offset on LightGCN (LightGCN-p), it gives +31.98 [20.65] on Coat ($p_{\mathrm{Holm}} = 0.013$ in the primary family of Table 6; 0.046 in the family of Table 7), and +21.22 [2.86] on KuaiRec ($p_{\mathrm{Holm}} < 0.001$). On Yahoo! R3 it gives −1.43 [3.93] ($p_{\mathrm{Holm}} = 0.475$), and on KuaiRand-Pure −0.81 [1.33] ($p_{\mathrm{Holm}} = 0.234$). The offset-only summary uses a different Holm family and reports 0.008 for Coat and 0.468 for KuaiRand-Pure. The offset-only gain on Coat is not robust to the Nadeau-Bengio correction ($p_{\mathrm{NB,Holm}} = 1.000$ in the primary family), and it is 0.061 over all comparisons with LightGCN (Table A.12). The offset helps on Coat and KuaiRec and not on Yahoo! R3 or KuaiRand-Pure. The tests here do not explain this pattern.

Table 7 separates the additive prior into its parts: the IPS weighting, the item term, and popularity. Removing the IPS weights does not reduce the gain. On KuaiRec, the unweighted prior added to iALS gives +22.79 [4.04] over iALS, and it exceeds the weighted prior by +8.05 [2.06] (p_{Holm} ¡ 0.001). On Coat the unweighted prior is above the weighted one by +6.27 [6.97] for iALS and +4.01 [10.01] for LightGCN, neither significant. Popularity does not reproduce the effect on KuaiRec: its offset adds +1.06 [2.46] to iALS ($p_{\mathrm{Holm}} = 1.000$), against +14.74 [3.43] for the weighted prior. Replacing the item term by a random permutation across items removes the KuaiRec gain and reverses it, with −112.07 [9.39] against iALS (p_{Holm} ¡ 0.001). On Coat the same permutation gives −14.06 [18.53] for iALS and −12.95 [24.18] for LightGCN, neither significant, so the Coat result does not isolate the item term. The lowest offset weight of the grid was selected in all five KuaiRec

splits for the shuffled item term, so the reported harm is an upper bound under that grid.

The training-free rows of Table 7 show the prior alone, with no learner. On KuaiRec the unweighted prior alone reaches 0.6367, and the weighted prior alone 0.6216. Popularity alone reaches 0.5766. On KuaiRand-Pure the picture is different: popularity alone (0.4204) exceeds the weighted prior alone (0.3637) by +56.41 [5.19] (p_{Holm} ¡ 0.001), and the unweighted prior alone (0.3837) exceeds the weighted one by +19.90 [2.45]. On Coat and Yahoo! R3, popularity alone is not significantly different from the weighted prior alone. The paired tests in the table are between the variants of the prior. No paired test of the prior alone against the training-free DRUP operator is in the result files, so the 0.0011 gap between the unweighted prior alone and DRUP on KuaiRec is reported as a difference of means only.

The weights on the sampled negatives form the second ingredient. On Yahoo! R3 the weights without the offset (DR-BPR-LGN-1) exceed LightGCN-p by +15.15 [5.84] (p_{Holm} ¡ 0.001). On Yahoo! R3 a shuffled Q removes the gain of the weighted learner over the backbone, with −1.01 [3.20] against LightGCN ($p_{\text{Holm}} = 0.534$), and popularity weights are significantly below LightGCN (−15.91 [6.06], $p_{\text{Holm}} < 0.001$). On Coat, shuffled Q gives +0.97 [5.50] against LightGCN ($p_{\text{Holm}} = 0.73$), while popularity weights give +6.91 [6.04] ($p_{\text{Holm}} = 0.0505$, not significant). The QP family exceeds LightGCN-p on KuaiRand-Pure by +4.74 [1.54] ($p_{\text{Holm}} < 0.001$), but the Nadeau-Bengio correction gives $p_{\text{NB,Holm}} = 0.054$ for this comparison, and no shuffled control was run on KuaiRand-Pure. In that dataset the validated combination selected $\beta = 0$ in four of five splits, so the gain there comes from the weights in those splits. The weight claim is therefore supported by the shuffled control on Yahoo! R3 and by the selection counts on KuaiRand-Pure, and it is not established on KuaiRec or Coat.

The refinements built on the prior do not improve on the simple prior of the same learner on Coat and KuaiRec. On Coat, the QP family is not different from LightGCN-p (+0.05 [8.29], TOST excludes differences larger than 0.010, p = 0.009). The largest refinement difference from LightGCN-p is +3.27 [6.94] (DR-BPR-LGN-p2), which is not significant. The Q-only variants are −8.02 [18.85] and −10.79 [19.26] below it. On iALS, DR-iALS is −10.22 [15.96] against iALS-p ($p_{\text{Holm}} = 0.208$). On KuaiRec the QP family is −0.32 [0.50] against LightGCN-p ($p_{\text{Holm}} = 0.211$, within ±0.003). On Yahoo! R3 and KuaiRand-Pure the QP family does exceed LightGCN-p, by +18.20 [5.64] and +4.74 [1.54] (both $p_{\text{Holm}} < 0.001$, Table 8), so the refinements are

not controls there. Every refinement in the study is below DRUP (0.6378) on KuaiRec. The DRGP refinement, which uses a leave-edge-out DR-fitted prior, reaches 0.6318, which is −5.58 [1.90] against DRUP (p_{Holm} ¡ 0.0001). These are single-run results with no seed repeats, and DRGP was run on Coat and KuaiRec only. Its Coat score of 0.5992 is −0.13 [15.11] against QP-LightGCN ($p_{\text{Holm}} = 1.000$). A validated ensemble of QP-LightGCN and DRUP selects one component in most splits and does not add to either: DRUP in four of five KuaiRec splits and QP-LightGCN in nine of ten Coat splits (`results/v4/ensemble_summary.txt`). On KuaiRec the ensemble is −0.18 [0.19] against DRUP ($p_{\text{Holm}} = 0.0664$), and on Coat it is +1.79 [2.39] against QP-LightGCN ($p_{\text{Holm}} = 0.1421$).

Two checks on the prior concern its inputs. The first is the propensity model. Coat's shipped propensities were estimated by the dataset authors from the uniform sample, so we refitted the prior rows with the estimator used on the other three datasets. Table 9 shows that this changes little. The estimated-minus-shipped difference is +2.63 [5.27] for iALS-p and +1.64 [10.30] for LightGCN-p, and the imputation alone changes by −1.14 [7.26]. None is significant. The gain over the no-prior reference remains as a point estimate: +19.44 [18.66] for iALS-p ($p_{\text{Holm}} = 0.330$, not significant) and +33.62 [16.72] for LightGCN-p. The estimated-propensity run of LightGCN was not carried out, and the other propensity-based Coat rows were not refitted.

The second check is the tuning budget of the backbone. The stored iALS grid selects its largest value of the confidence weight α (30) in all 10 splits, and the wider grid of Table 10 selects its largest value (1000) in four splits. The stored grid is therefore bounded at its edge, and the wider grid is not exhausted. Over the stored iALS, the wider grid gives +12.92 [8.01] ($p_{\text{Holm}} = 0.003$), against +16.82 [18.04] for the offset model over the same stored iALS. The wider grid is not significantly different from iALS-p (−3.90 [15.99], $p_{\text{Holm}} = 0.631$). The two differences cannot be separated: the wider grid has 32 configurations against 90 for iALS-p, and the gap between iALS and the prior-augmented model on Coat is not resolved by the present tests.

The stored Yahoo! R3 DR-LightGCN row (0.5528, 6 configurations) lies far below the other graph models. It is one of the three replay exceptions of Section 3.4, and we cannot tell whether under-tuning or an implementation limit causes this. The Impute row on KuaiRec (0.6194, Table 4) differs from the prior alone (0.6216, Table 7) in its propensity floor and penalty. The Impute row is produced by the training-free runner, which fixes the floor of

the imputation fit at 0.05 and selects the penalty $\lambda = 20$ from $\{5, 20\}$. The prior alone uses floor 0.2 and penalty $\lambda = 5$. The two rows are therefore not separate fits of the same model.

5.4. Exact unbiasedness of the propagation and the training-free results (RQ4)

The walk correction removes the repeated-edge terms of multi-hop doubly robust propagation. On a synthetic MNAR model (`results/v2/mc_stress_K3_p1.0.json` and `results/v2/mc_stress_K5_p1.0.json`, independent exposure), it reduces the relative |bias| of the DR adjacency from 1.756 to 0.019 at $K = 3$ and from 8.350 to 0.039 at $K = 5$. On a semi-synthetic KuaiRec log built from the dense setting (`results/v3/semisynth_kuairec_dense.json`; 30 replications; $\beta = 1$; true propensities; fixed degree weights), the relative bias of the three-hop term is 1.95 for DR and 0.88 for DRUP, and 3.46 against 0.32 at $K = 5$. With estimated propensities the same comparison gives 9.47 against 5.66. In the three cells with degree weights computed from the imputation (true propensities), the walk correction changes the relative bias from 0.67 to 0.66 (independent), from 0.72 to 0.75 (cross-fitted) and from 0.53 to 0.32 (split). The stored semi-synthetic runs measure relative bias of the three-hop term at $\beta = 1$ only. They do not contain nDCG values, so the effect of the correction on ranking in that setting is not measured.

On the real logs the correction does not help at matched configurations. On Coat, DRUP minus DR is negative at every value of β in the grid, from -0.33 at $\beta = 0.03$ to -8.17 at $\beta = 1000$, and it is negative in all ten splits for $\beta \geq 0.3$. On Yahoo! R3 the differences are small and negative, from -0.02 to -0.32. Among the selected runs, the difference is $+1.40$ [3.86] on Coat ($p_{\text{Holm}} = 0.475$), and the data exclude differences larger than 0.010 (TOST p ¡ 0.001). On KuaiRec the selected-run difference is -0.12 [0.15] ($p_{\text{Holm}} = 0.113$), within ± 0.003 (TOST p ¡ 0.001). Table 11 gives the matched-configuration and selected-run values. Exact unbiasedness therefore does not improve ranking on the three real logs on which DR and DRUP were run. DR and DRUP were not run on KuaiRand-Pure.

Among the training-free operators, the best KuaiRec mean goes to doubly robust propagation, not to the walk correction. DRUP reaches 0.6378 and the DR adjacency on which it is built also reaches 0.6378, and a five-hop DR adjacency reaches 0.6379 (`results/v2/filters_kuairec_pop.json`). No trained model in Table 4 exceeds 0.6378 on KuaiRec: the best is DR-JL at 0.6332, and the QP family reaches 0.6285. On Coat, DRUP is weak at 0.5467, and on Yahoo! R3 the DRUP and DR rows coincide at 0.6567. In

the bootstrap, DR ranks first in 87.3% of resamples on KuaiRec and DRUP in 12.7% (Figure 1), which reflects the tie in their means. The unweighted prior alone (0.6367) is the one training-free result close to DRUP, and it is reported in Table 7. The reason the walk correction does not change the ranking on these logs is not identified by our runs.

6. Discussion

6.1. What the evidence supports

Under a protocol that keeps uniform data out of training, apart from Coat's shipped propensities, and selects on validation data, the ranking of debiasing methods depends on evaluation choices in ways that matter at the scale of the differences being compared. Selecting on test data inflates the Coat scores by up to 0.030 nDCG@5, and under the per-split test oracle it reverses the order of iALS and LightGCN. The differences among the leading Coat methods are of the same order (Table 4), so a ranking of those methods made on test data is not reliable without the validation step.

Third-party graph-specific debiasing adds little over an equally tuned backbone on Coat, Yahoo! R3 and KuaiRand-Pure. The largest point-estimate gain over LightGCN on these three datasets is +4.10 ($\times 10^{-3}$) nDCG, for r-AdjNorm on Yahoo! R3, and the Nadeau-Bengio correction removes its significance. On Coat the 95% half-widths of the debiasers reach ±30, so the study cannot exclude gains of the size of the minimum detectable difference. Our own QP-LightGCN family is larger on Yahoo! R3 (+16.77 [5.82] over LightGCN) and on KuaiRand-Pure (+3.93 [1.53]). Its gains are not significant under the Nadeau-Bengio correction against LightGCN, and the refinement comparisons that the family supports are in Section 5.3. KuaiRec is different. NAVIP and DR-LightGCN add +10.41 and +13.17 ($\times 10^{-3}$) there, although r-AdjNorm adds only +0.22, and a change of training loss without any debiasing gives a larger mean difference, +19.59. A reader comparing debiasers on KuaiRec therefore needs the pointwise backbone as the reference, not only the BPR one. The two differences are not compared by a paired test, so the pointwise result is a difference of means.

On Coat and KuaiRec the additive prior is the ingredient with the largest point-estimate effect over LightGCN that we tested. Added as a score offset to LightGCN, it gives +31.98 [20.65] on Coat and +21.22 [2.86] on KuaiRec, and it does not change Yahoo! R3 or KuaiRand-Pure. The Coat gain is not robust to the Nadeau-Bengio correction ($p_{\mathrm{NB,Holm}} = 1.000$ in the primary

family), and it is 0.061 over all comparisons with LightGCN. The KuaiRec gain is robust to that correction ($p_{\mathrm{NB,Holm}} = 0.009$). The evidence does not show that the inverse-propensity weighting matters: on KuaiRec the unweighted prior is significantly better than the weighted one, and on Coat it is numerically higher without a significant difference. A random permutation of the item term removes the KuaiRec gain, but on Coat the same permutation does not significantly change the score, so the item term is established on KuaiRec and not on Coat. These results do not establish why the prior helps. The reading that the item term uses exposed negatives that a BPR objective discards is a hypothesis, and the prior-alone comparisons do not test it.

The weights on the sampled negatives are a second ingredient, but their support differs by dataset. On Yahoo! R3 a shuffled weight vector removes the gain of the weighted learner over the backbone, and popularity weights are significantly below LightGCN, so the weights are not a popularity reweighting. On KuaiRand-Pure the QP family exceeds LightGCN-p by +4.74 [1.54], but no shuffled control was run there and the Nadeau-Bengio correction gives $p_{\mathrm{NB,Holm}} = 0.054$ for that comparison. On KuaiRand-Pure, moreover, the popularity prior alone outperforms the IPS-fitted prior alone by +56.41 [5.19], so the simple non-personalised baseline is a serious competitor there, and a study that omits it would overstate the debiased prior.

On KuaiRec, training-free doubly robust propagation has the highest mean in Table 4, and no trained method or refinement in the study exceeds it. This is a difference of means: DR and DRUP are tied, and the bootstrap cannot separate them. The walk correction that makes its propagation exactly unbiased on synthetic logs does not change accuracy on the three real logs on which DR and DRUP were run (Coat, Yahoo! R3 and KuaiRec). These data do not show what limits the training-free propagation, and KuaiRand-Pure has no DR or DRUP rows.

Finally, the backbone matters as much as the debiaser. A wider iALS grid on Coat raises the stored iALS by +12.92 [8.01] ($p_{\mathrm{Holm}} = 0.003$), against +16.82 [18.04] for the offset model over the same stored iALS. The wider grid is 32 configurations against 90 for iALS-p, and the present tests cannot separate tuning budget from the prior on Coat. An evaluation that tunes the proposed method more than its baselines can produce a gap of that size from budget alone.

6.2. Implications for evaluating debiasing methods

The results support the following practices. Each item rests on a result above.

1. **Keep uniform data out of training, and state where every nuisance comes from.** Coat's shipped propensities came from the uniform sample. Refitting them with the estimator used on the other datasets changes the Coat prior results by at most +2.63 [5.27] (Table 9). The check is cheap, and it should be reported whenever the propensities are not fitted on the training log. It covers only the prior rows of the study.
2. **Select per split on validation data, and report the test-oracle gap.** The gap is +0.0097 to +0.0302 on Coat, and under the per-split test oracle it reverses the order of iALS and LightGCN there (Section 5.1).
3. **Report the budget, and compare at matched budget.** Grid sizes in these experiments range from 6 to 324 configurations. The sign of the DRUP comparison with DR-JL on KuaiRec changes in 78.4% of six-configuration subgrids (Table 5).
4. **Include the simple baselines as mandatory rows:** the IPS imputation alone, a tuned iALS with its wider grid, pointwise LightGCN, the popularity ranker, and the backbone with the debiased prior. Several of these changed a conclusion here: pointwise LightGCN on KuaiRec (+19.59 [4.55] over the BPR backbone, as a difference of means), popularity against the prior alone on KuaiRand-Pure, and the wider iALS grid on Coat.
5. **Retrain the selected configurations with several seeds.** On Coat, the difference between QP-LightGCN and iALS is +20.02 [19.74] ($p_{\text{Holm}} = 0.0469$) in the main run and +3.39 [20.70] ($p_{\text{Holm}} = 0.747$) over seeds 1 to 3. Both comparisons are descriptive, because the baselines were chosen on test data.
6. **Report the minimum detectable difference with every claim of no difference, and test equivalence with a margin fixed in advance.** On Coat the minimum detectable difference for the primary comparisons is 0.012 to 0.043, which is large relative to the method differences in Table 4. The MDE ignores the dependence between splits, so it is optimistic.

7. **Use a correction for overlapping splits, and report it next to the standard test.** Under the Nadeau-Bengio correction, no graph-debiaser comparison (P1) is significant on any dataset. The correction used here is heuristic and conservative for five splits (Section 3.6).
8. **Publish the per-split validation and test results.** The selection analyses here, including the inflation, the subgrid stability and the bootstrap ranks, can be redone only from per-split values.

6.3. Limitations and threats to validity

1. **Power on Coat.** Coat has 225 evaluated users. The minimum detectable differences of the primary comparisons range from 0.012 to 0.043. Coat results are therefore statements of the form "no detectable difference, and differences larger than the stated margin are excluded", never statements of equivalence.
2. **Sensitivity margins.** The TOST margins (± 0.010 on Coat and Yahoo! R3, ± 0.003 on KuaiRand-Pure and KuaiRec) are sensitivity thresholds chosen before the analysis. They are not derived from the literature, and the TOST results should be read as a sensitivity analysis of those thresholds.
3. **Our implementations.** The graph baselines NAVIP, r-AdjNorm, SimGCL and MACR, and the refinements, are our implementations. AdvInfoNCE is a simplified re-implementation: its adversary budget is implicit, it has no KL-ball projection, and its grid differs across datasets. The official AdvInfoNCE code and the PPD post-hoc method were not run. The negative results for these methods are therefore about our implementations under this protocol. The Yahoo! R3 DR-LightGCN row (0.5528, 6 configurations) is far below the other graph models, and it may reflect under-tuning or an implementation limit.
4. **The refinement family is ours.** The refinements of the prior (Section 4) were developed as candidate methods, and they are reported in full with their configuration counts. The family is not exhaustive, and the results do not show that no other refinement helps.
5. **Seeds are incomplete.** Seeds 1 to 3 were not run for the Coat prior rows. Seed checks cover the leading methods only, and their baselines were chosen on test data, so those comparisons are descriptive. DRGP was run on Coat and KuaiRec only, and its KuaiRec and Coat results are single runs without seeds.

6. **Budgets are unequal and only partly matched.** Matched-budget differences are split means without tests. The sensitivity analysis draws random subgrids of the stored configurations, not new searches.
7. **Grid edges.** On KuaiRec, the popularity and shuffled priors select the lowest offset weight of their grid in all five splits, so the reported shuffled-prior harm is an upper bound under that grid. On KuaiRec the unweighted prior selects the largest regularisation value of its grid (1000). The wider iALS grid is not exhausted.
8. **Incomplete ablations.** The user-only prior was run only as a prior alone and not in a learner. LightGCN with estimated propensities was not run. On KuaiRec, only the additive prior was run in the offset-only LightGCN setting.
9. **Replay gaps.** Replaying each stored selection from its bank reproduces 151 of 154 method means within 0.002. Three do not. On replay the KuaiRec DR-LightGCN mean is 0.0033 higher than the stored 0.6212, the Yahoo! R3 DR-LightGCN mean 0.0058 higher, and the Yahoo! R3 MACR mean 0.0060 higher. The stored values are reproduced only with an early-stopping setting that the bank does not record. The stored values are used in Table 4, and the KuaiRec value also enters the P1 comparison of Table 6.
10. **Family dependence.** Several p-values change with the Holm family. The Yahoo! R3 r-AdjNorm comparison has $p_{\text{Holm}} = 0.029$ in the primary family and 0.073 over all comparisons with LightGCN. The Coat offset comparison has 0.013 in the primary family, 0.046 in the family of the prior-anatomy table and 0.061 over all comparisons with LightGCN. We report the family with each value.
11. **Overlapping splits.** Per-user tests over overlapping splits treat the log and the fitted nuisances as fixed, and they can be anti-conservative. The Nadeau-Bengio correction used here is a heuristic (Section 3.6), and it removes significance from every P1 comparison.
12. **Exposure semantics.** On Coat and Yahoo! R3, exposure is user self-selection. The derivation of the weights assumes that exposure is independent of the outcome given the covariates of the propensity model.
13. **Test design.** We report one K per dataset in the main text (nDCG@5 on Coat and Yahoo! R3, nDCG@10 on KuaiRand-Pure, nDCG@20 on KuaiRec). The alternative metric is computed on the stored selections

only. KuaiRec has a fully observed test matrix with about 3,300 candidates per user, which differs from the other three datasets, and we have no test of which feature causes the difference.

14. **Unexplained differences.** The Impute row and the prior-alone row on KuaiRec differ in propensity floor and penalty (Section 5.3). No paired test compares the prior alone with DRUP, so the 0.0011 gap to DRUP is a difference of means.
15. **Synthetic evidence for the walk correction.** The bias results come from a synthetic MNAR model and a semi-synthetic KuaiRec study that reports relative bias of the three-hop term at one value of β. Neither measures ranking, and their bias reductions do not carry over to the real logs at matched configurations.
16. **The mechanism is untested.** The item-term reading in Section 6.1 is a hypothesis. The results show that an item-level term fitted once on the exposed log carries the effect on KuaiRec. They do not show why.

7. Conclusion

We re-evaluated exposure-bias corrections for graph recommenders on four benchmarks with uniformly exposed test data, under one protocol in which uniform data serve for validation and testing, and, through Coat's shipped propensities, for one nuisance that we check for the prior rows. The study asked how much evaluation choices move conclusions, how much graph-specific debiasing adds over an equally tuned backbone, which ingredient carries the gains that remain, and whether exact unbiasedness of the propagation matters for ranking.

Four findings stand out. First, the protocol is sensitive to evaluation choices. Selecting on test data inflates Coat scores by up to 0.030 nDCG@5, and under the per-split test oracle it reverses the order of iALS and LightGCN. A wider iALS grid on Coat raises iALS by +12.92 ($\times 10^{-3}$), so tuning budget alone can produce gaps of the same order as the differences being compared. Second, third-party graph-specific debiasers add at most +4.10 ($\times 10^{-3}$) nDCG over LightGCN in the point estimates on Coat, Yahoo! R3 and KuaiRand-Pure, and none of these gains survives the correction for overlapping splits. On KuaiRec the gains are larger, but a pointwise loss without any debiasing gives a larger mean difference. Our own QP-LightGCN family adds +16.77 on Yahoo! R3 and +3.93 on KuaiRand-Pure over LightGCN,

and these differences are not significant under the same correction. Third, an additive prior, fitted once on the exposed log and added to the score, gives the largest point-estimate gains over LightGCN on Coat and KuaiRec. The Coat gain is not robust to the correction for overlapping splits. Removing the inverse-propensity weights does not reduce the gain, a shuffled item term removes the KuaiRec gain, and popularity does not reproduce it. The mechanism is not identified. Fourth, the walk correction, which makes multi-hop doubly robust propagation exactly unbiased, does not improve ranking on Coat, Yahoo! R3 or KuaiRec. DR and DRUP were not run on KuaiRand-Pure.

The study does not propose a method. It does not show that debiasing is ineffective in general, because its scope is four public benchmarks, one cut-off per dataset and offline evaluation of implicit-feedback learners trained on the biased log. It does not establish equivalence on Coat, where the study is underpowered. For evaluation, the practical requirements are to select per split on validation data, to match budgets, to include the simple baselines, and to report minimum detectable effects and corrected tests with every claim that two methods are equal or that one is better.

Appendix A. Every method against the backbone

Tables A.12 to A.15 give, for each dataset, every method against LightGCN, which is the backbone family of Section 3.6.

References

Anelli, V.W., Malitesta, D., Pomo, C., Bellogin, A., Di Sciascio, E., Di Noia, T., 2023. Challenging the myth of graph collaborative filtering: a reasoned and reproducibility-driven analysis, in: Proceedings of the 17th ACM Conference on Recommender Systems, ACM. pp. 350–361. doi:10.1145/3604915.3609489.

Bonner, S., Vasile, F., 2018. Causal embeddings for recommendation, in: Proceedings of the 12th ACM Conference on Recommender Systems, pp. 104–112. doi:10.1145/3240323.3240360.

Chen, C., Lin, H., Xu, Y., 2026. Towards reliable negative sampling for recommendation with implicit feedback via in-community popularity. arXiv:2602.18759. preprint.

Chen, J., Dong, H., Qiu, Y., He, X., Xin, X., Chen, L., Lin, G., Yang, K., 2021. AutoDebias: Learning to debias for recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 21–30. doi:10.1145/3404835.3462919.

Chen, J., Dong, H., Wang, X., Feng, F., Wang, M., He, X., 2023. Bias and debias in recommender system: A survey and future directions. ACM Transactions on Information Systems 41, 1–39. doi:10.1145/3564284.

Chen, J., Wu, J., Chen, J., Xin, X., Li, Y., He, X., 2024. How graph convolutions amplify popularity bias for recommendation? Frontiers of Computer Science 18. doi:10.1007/s11704-023-2655-2.

Choi, J., Hong, S., Park, N., Cho, S.B., 2023. Blurring-sharpening process models for collaborative filtering, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 1096–1106. doi:10.1145/3539618.3591645.

Chuang, C.Y., Robinson, J., Lin, Y.C., Torralba, A., Jegelka, S., 2020. Debiased contrastive learning, in: Advances in Neural Information Processing Systems 33 (NeurIPS 2020). URL: https://papers.nips.cc/paper/2020/hash/63c3ddcc7b23daa1e42dc41f9a44a873-Abstract.html, arXiv:2007.00224.

Dai, Q., Li, H., Wu, P., Dong, Z., Zhou, X.H., Zhang, R., Zhang, R., Sun, J., 2022. A generalized doubly robust learning framework for debiasing post-click conversion rate prediction, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, ACM. pp. 252–262. doi:10.1145/3534678.3539270.

Ding, J., Quan, Y., Yao, Q., Li, Y., Jin, D., 2020. Simplify and robustify negative sampling for implicit collaborative filtering, in: Advances in Neural Information Processing Systems 33 (NeurIPS 2020). URL: https://papers.nips.cc/paper/2020/hash/0c7119e3a6a2209da6a5b90e5b5b75bd-Abstract.html, arXiv:2009.03376.

Dodge, J., Gururangan, S., Card, D., Schwartz, R., Smith, N.A., 2019. Show your work: Improved reporting of experimental results, in: Proceedings of

the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 2185–2194. doi:10.18653/v1/D19-1224.

Ferrari Dacrema, M., Cremonesi, P., Jannach, D., 2019. Are we really making much progress? A worrying analysis of recent neural recommendation approaches, in: Proceedings of the 13th ACM Conference on Recommender Systems, ACM. pp. 101–109. doi:10.1145/3298689.3347058.

Gao, C., Li, S., Lei, W., Chen, J., Li, B., Jiang, P., He, X., Mao, J., Chua, T.S., 2022a. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems [data set], in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 540–550. doi:10.1145/3511808.3557220.

Gao, C., Li, S., Zhang, Y., Chen, J., Li, B., Lei, W., Jiang, P., He, X., 2022b. KuaiRand: An unbiased sequential recommendation dataset with randomly exposed videos [data set], in: Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM), pp. 3953–3957. doi:10.1145/3511808.3557624.

Gao, C., Zheng, Y., Wang, W., Feng, F., He, X., Li, Y., 2024. Causal inference in recommender systems: A survey and future directions. ACM Transactions on Information Systems 42, 1–32. doi:10.1145/3639048.

Guo, S., Zou, L., Liu, Y., Ye, W., Cheng, S., Wang, S., Chen, H., Yin, D., Chang, Y., 2021. Enhanced doubly robust learning for debiasing post-click conversion rate estimation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 275–284. doi:10.1145/3404835.3462917.

He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., Wang, M., 2020. LightGCN: Simplifying and powering graph convolution network for recommendation, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 639–648.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70. URL: https://www.jstor.org/stable/4615733.

Hu, Y., Koren, Y., Volinsky, C., 2008. Collaborative filtering for implicit feedback datasets, in: 2008 Eighth IEEE International Conference on Data Mining (ICDM), pp. 263–272. doi:10.1109/ICDM.2008.22.

Islam, M.A., Faruk, A.S., Medya, S., Zheleva, E., 2026a. Debiasing message passing to mitigate popularity bias in GNN-based collaborative filtering. arXiv:2605.11145. preprint.

Islam, M.A., Zheleva, E., Wang, R., 2026b. Post-hoc popularity bias correction in GNN-based collaborative filtering, in: Proceedings of the ACM Web Conference 2026 (WWW '26), Dubai, United Arab Emirates. doi:10.1145/3774904.3792670, arXiv:2510.12959.

Kim, M., Oh, J., Do, J., Lee, S., 2022. Debiasing neighbor aggregation for graph neural network in recommender systems, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 4128–4132. doi:10.1145/3511808.3557576.

Krichene, W., Rendle, S., 2020. On sampled metrics for item recommendation, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, ACM. pp. 1748–1757. doi:10.1145/3394486.3403226.

Kweon, W., Yu, H., 2024. Doubly calibrated estimator for recommendation on data missing not at random, in: Proceedings of the ACM Web Conference 2024 (WWW '24), Singapore. doi:10.1145/3589334.3645417, arXiv:2403.00817.

Lee, J.w., Park, S., Lee, J., Lee, J., 2022. Bilateral self-unbiased learning from biased implicit feedback, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '22), Madrid, Spain. arXiv:2207.12660.

Li, H., Zheng, C., Wu, P., 2023. StableDR: Stabilized doubly robust learning for recommendation on data missing not at random, in: International Conference on Learning Representations (ICLR).

Liang, D., Charlin, L., McInerney, J., Blei, D.M., 2016. Modeling user exposure in recommendation, in: Proceedings of the 25th International Conference on World Wide Web, pp. 951–961. doi:10.1145/2872427.2883090.

Liu, D., Cheng, P., Dong, Z., He, X., Pan, W., Ming, Z., 2020. A general knowledge distillation framework for counterfactual recommendation via uniform data, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20), pp. 831–840. doi:10.1145/3397271.3401083. full text not read; bibliographic data from the ACM reference format quoted by a secondary source. UNVERIFIED against the proceedings.

Mao, K., Zhu, J., Xiao, X., Lu, B., Wang, Z., He, X., 2021. UltraGCN: Ultra simplification of graph convolutional networks for recommendation, in: Proceedings of the 30th ACM International Conference on Information and Knowledge Management, pp. 1253–1262. doi:10.1145/3459637.3482291.

Marlin, B.M., Zemel, R.S., 2009. Collaborative prediction and ranking with non-random missing data [data set], in: Proceedings of the Third ACM Conference on Recommender Systems (RecSys), pp. 5–12. doi:10.1145/1639714.1639717.

Menon, A.K., Jayasumana, S., Rawat, A.S., Jain, H., Veit, A., Kumar, S., 2021. Long-tail learning via logit adjustment, in: International Conference on Learning Representations (ICLR). arXiv:2007.07314.

Nadeau, C., Bengio, Y., 2003. Inference for the generalization error. Machine Learning 52, 239–281. doi:10.1023/a:1024068626366.

Que, Y., Zhang, Y., Zhao, X., Ma, C., 2025. Causality-aware graph aggregation weight estimator for popularity debiasing in top-k recommendation, in: Proceedings of the 34th ACM International Conference on Information and Knowledge Management, ACM. pp. 2471–2481. doi:10.1145/3746252.3761155.

Raja, R., Vats, A., 2025. Counterfactual risk minimization with IPS-weighted BPR and self-normalized evaluation in recommender systems, in: Causality, Counterfactuals and Sequential Decision-Making Workshop at RecSys 2025. arXiv:2509.00333. workshop paper.

Rendle, S., Freudenthaler, C., Gantner, Z., Schmidt-Thieme, L., 2009. BPR: Bayesian personalized ranking from implicit feedback, in: Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI), pp. 452–461. arXiv:1205.2618.

Rendle, S., Krichene, W., Zhang, L., Koren, Y., 2022. Revisiting the performance of iALS on item recommendation benchmarks, in: Proceedings of the 16th ACM Conference on Recommender Systems, ACM. pp. 427–435. doi:10.1145/3523227.3548486, arXiv:2110.14037.

Rendle, S., Zhang, L., Koren, Y., 2019. On the difficulty of evaluating baselines: A study on recommender systems. arXiv:1905.01395. preprint.

Saito, Y., 2020. Unbiased pairwise learning from biased implicit feedback, in: Proceedings of the 2020 ACM SIGIR on International Conference on Theory of Information Retrieval (ICTIR), ACM. pp. 5–12. doi:10.1145/3409256.3409812.

Saito, Y., Yaginuma, S., Nishino, Y., Sakata, H., Nakata, K., 2020. Unbiased recommender learning from missing-not-at-random implicit feedback, in: Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM), pp. 501–509.

Schnabel, T., Swaminathan, A., Singh, A., Chandak, N., Joachims, T., 2016. Recommendations as treatments: Debiasing learning and evaluation [data set], in: Proceedings of the 33rd International Conference on Machine Learning (ICML), pp. 1670–1679.

Schuirmann, D.J., 1987. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. Journal of Pharmacokinetics and Biopharmaceutics 15, 657–680. doi:10.1007/BF01068419.

Shen, Y., Wu, Y., Zhang, Y., Shan, C., Zhang, J., Letaief, K.B., Li, D., 2021. How powerful is graph convolution for recommendation?, in: Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM), pp. 1619–1629.

Song, Z., Chen, J., Zhou, S., Shi, Q., Feng, Y., Chen, C., Wang, C., 2023. CDR: Conservative doubly robust learning for debiased recommendation, in: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23), Birmingham, United Kingdom. doi:10.1145/3583780.3614805, arXiv:2308.08461.

Steck, H., 2010. Training and testing of recommender systems on data missing not at random, in: Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM. pp. 713–722. doi:10.1145/1835804.1835895.

Steck, H., 2019. Embarrassingly shallow autoencoders for sparse data, in: The World Wide Web Conference, ACM. pp. 3251–3257. doi:10.1145/3308558.3313710.

Swaminathan, A., Joachims, T., 2015. The self-normalized estimator for counterfactual learning, in: Advances in Neural Information Processing Systems, pp. 3231–3239.

Wang, B., Chen, J., Li, C., Zhou, S., Shi, Q., Gao, Y., Feng, Y., Chen, C., Wang, C., 2024. Distributionally robust graph-based recommendation system, in: Proceedings of the ACM Web Conference 2024 (WWW '24), Singapore. doi:10.1145/3589334.3645598, arXiv:2402.12994.

Wang, X., He, X., Wang, M., Feng, F., Chua, T.S., 2019a. Neural graph collaborative filtering, in: Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 165–174. doi:10.1145/3331184.3331267.

Wang, X., Zhang, R., Sun, Y., Qi, J., 2019b. Doubly robust joint learning for recommendation on data missing not at random, in: Proceedings of the 36th International Conference on Machine Learning (ICML), pp. 6638–6647.

Wang, Z., Chen, X., Wen, R., Huang, S.L., Kuruoglu, E.E., Zheng, Y., 2020. Information theoretic counterfactual learning from missing-not-at-random feedback, in: Advances in Neural Information Processing Systems.

Wei, T., Feng, F., Chen, J., Wu, Z., Yi, J., He, X., 2021. Model-agnostic counterfactual reasoning for eliminating popularity bias in recommender system, in: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 1791–1800. doi:10.1145/3447548.3467289.

Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., Xie, X., 2021. Self-supervised graph learning for recommendation, in: Proceedings of the

44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 726–735. doi:10.1145/3404835.3462862.

Wu, S., Sun, F., Zhang, W., Xie, X., Cui, B., 2023. Graph neural networks in recommender systems: A survey. ACM Computing Surveys 55, 1–37. doi:10.1145/3535101.

Yin, M., Hu, C., Liu, B., Gong, N.Z., Ye, Y., Li, X., 2026. Correct and weight: A simple yet effective loss for implicit feedback recommendation. arXiv:2601.04291. preprint.

Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., Nguyen, Q.V.H., 2022. Are graph augmentations necessary?: Simple graph contrastive learning for recommendation, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 1294–1303. doi:10.1145/3477495.3531937.

Zhang, A., Ma, W., Wang, X., Chua, T.S., 2022. Incorporating bias-aware margins into contrastive loss for collaborative filtering, in: Advances in Neural Information Processing Systems 35 (NeurIPS 2022). URL: https://proceedings.nips.cc/paper_files/paper/2022/file/334da4cbb76302f37bd2e9d86f558869-Paper-Conference.pdf, arXiv:2210.11054.

Zhang, A., Ma, W., Wei, P., Sheng, L., Wang, X., 2024. General debiasing for graph-based collaborative filtering via adversarial graph dropout, in: Proceedings of the ACM Web Conference 2024 (WWW '24), Singapore. doi:10.1145/3589334.3645667, arXiv:2402.13769.

Zhang, A., Sheng, L., Cai, Z., Wang, X., Chua, T.S., 2023. Empowering collaborative filtering with principled adversarial contrastive loss, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023). URL: https://neurips.cc/virtual/2023/poster/71149, arXiv:2310.18700.

Zhang, J., Wang, Y., Song, Q., Hong, L., 2026. ConAlign: Conditional alignment framework for balancing biased and unbiased recommendation. arXiv:2607.24092. preprint.

Zhang, Y., Feng, F., He, X., Wei, T., Song, C., Ling, G., Zhang, Y., 2021. Causal intervention for leveraging popularity bias in recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 11–20. doi:10.1145/3404835.3462875.

Zhao, M., Wu, L., Liang, Y., Chen, L., Zhang, J., Deng, Q., Wang, K., Shen, X., Lv, T., Wu, R., 2022. Investigating accuracy-novelty performance for graph-based collaborative filtering, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 50–59. doi:10.1145/3477495.3532005.

Zhou, H., Chen, H., Dong, J., Zha, D., Zhou, C., Huang, X., 2023. Adaptive popularity debiasing aggregator for graph collaborative filtering, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 7–17. doi:10.1145/3539618.3591635.

Table 1: Positioning of related studies by their protocol. Entries follow the protocols as described in the cited papers; "not stated" marks a point that the description does not settle. Uniform data are the uniformly collected or randomised samples of the benchmark.

Study	Graph backbone	Propensity or imputation	Uniform data in training	Uniform data for validation or selection	Unbiased test benchmarks
AutoDebias (Chen et al., 2021)	no (matrix factorisation)	learned weights and imputation	yes, small sample	yes, small sample	Coat, Yahoo! R3
CausE (Bonner and Vasile, 2018)	no (matrix factorisation)	none	yes, randomised sample	no, validation from logged policy	MovieLens, Netflix
CDR (Song et al., 2023)	no (matrix factorisation)	doubly robust family	no	yes, part of unbiased data	Coat, Yahoo! R3, KuaiRand
DCE (Kweon and Yu, 2024)	no (MF or NCF)	doubly robust loss	no	no, held out from biased training data	Coat, Yahoo! R3, KuaiRec
BISER (Lee et al., 2022)	no	self-inverse propensity	no	no, held out from biased training data	Coat, Yahoo! R3
ConAlign (Zhang et al., 2026)	no (dual tower)	MF-IPS and MF-DR baselines	yes	yes	Coat, Yahoo! R3, KuaiRand-Pure
NAVIP (Kim et al., 2022)	yes	inverse propensity on edges	no	pseudo-unbiased popularity sample	Yahoo! R3 and pseudo-unbiased sets
AdvDrop (Zhang et al., 2024)	yes	IPS and DR baselines only	no, for its own method	not stated	Coat, Yahoo!, KuaiRec
PPD (Islam et al., 2026b)	yes, post hoc	none	no	yes on Coat and KuaiRec; biased sample on Yahoo! R3	Coat, Yahoo! R3, KuaiRec
DPAA (Islam et al., 2026a)	yes	none	no	validation present, construction not confirmed	Coat, Yahoo! R3, KuaiRec
DR-GNN (Wang et al., 2024)	yes	none	not stated	not stated	Coat, Yahoo
This study	yes	IPS-fitted imputation and propensity	no, except Coat shipped propensities	yes	Coat, Yahoo! R3, KuaiRand-Pure, KuaiRec

Table 2: Permitted uses of each data source. The Coat row for the propensity is the one exception to the rule, and it is discussed in the text.

Stage	Log	Validation	Test
Exposure propensity $\hat{p}$ (Coat: shipped, see text)	yes	no	no
Additive imputation $\hat{Y}$, prior z, weights Q	yes	no	no
Cross-fitted nuisances of the DRGP prior	yes	no	no
Training of every model	yes	no	no
Choice of configuration and epoch within a split	no	yes	no
Early stopping within a split	no	yes	no
Reported test score of the chosen model	no	no	yes

Table 3: Datasets. "Log": the biased training data, with its density. "Test": the unbiased data. "Candidates": the unbiased test items per user that are ranked, which are the items never logged for that user. "Evaluated users": users with at least one positive candidate in the full test set. The analysis uses 225, 2,310, 8,321 and 1,410 users (Coat, Yahoo! R3, KuaiRand-Pure, KuaiRec), which differ from the evaluated users of 225, 2,315, 8,347 and 1,411 as explained in Section 3.2.

Dataset	Users / items	Log (density)	O means	Test	Candidates	Evaluated users
Coat	290 / 300	6,960 (8.0%)	Self-rated	16 random ratings per user	14.7	225
Yahoo! R3	15,400 / 1,000	311,704 (2.0%)	Self-rated	10 random songs, 5,400 users	10	2,315
KuaiRand	10,000 / 7,583	511,863 (0.68%)	Feed impression	random impressions	43.6	8,347
KuaiRec	7,176 / 10,728	10.3M (13.4%)	Impression	fully observed 1,411 × 3,327	3,314	1,411

Table 4: Main benchmark: mean test score over splits (sd over splits, ddof 0) and configuration count per method. Metric: nDCG@5 (Coat, Yahoo), nDCG@10 (KuaiRand), nDCG@20 (KuaiRec).

Method	Coat		Yahoo		KuaiRand		KuaiRec	
	mean (sd)	cfg	mean (sd)	cfg	mean (sd)	cfg	mean (sd)	cfg
Obs	0.5571 (0.0100)	24	0.6585 (0.0058)	16	–	–	0.6036 (0.0028)	16
Pop	0.5367 (0.0095)	1	0.5331 (0.0059)	1	–	–	0.5766 (0.0023)	1
Impute	0.5532 (0.0097)	6	0.5162 (0.0044)	2	–	–	0.6194 (0.0018)	4
IPS	0.5510 (0.0224)	720	–	–	–	–	–	–
DR	0.5459 (0.0244)	1440	0.6567 (0.0063)	512	–	–	0.6378 (0.0020)	384
DRUP	0.5467 (0.0235)	1440	0.6567 (0.0063)	512	–	–	0.6378 (0.0019)	384
EASE	0.5401 (0.0138)	3	–	–	–	–	–	–
GF-CF	0.5512 (0.0124)	4	–	–	–	–	–	–
EASE-DR	0.5564 (0.0141)	90	–	–	–	–	–	–
GF-CF-DR	0.5598 (0.0084)	120	–	–	–	–	–	–
MF	0.4780 (0.0168)	18	0.5087 (0.0075)	9	0.3705 (0.0008)	6	0.6221 (0.0036)	6
IPS-MF	0.4713 (0.0160)	18	0.5125 (0.0059)	9	–	–	0.6252 (0.0018)	6
DR-MF	0.5237 (0.0111)	18	0.5093 (0.0042)	9	0.3608 (0.0067)	6	0.6219 (0.0025)	6
BPR-MF	0.5174 (0.0206)	18	0.6461 (0.0042)	9	0.4475 (0.0027)	6	0.5963 (0.0028)	6
PDA	0.5181 (0.0138)	36	0.6514 (0.0051)	18	–	–	–	–
LightGCN	0.5683 (0.0132)	54	0.6708 (0.0061)	36	0.4569 (0.0010)	6	0.6085 (0.0028)	36
LightGCN-pt	0.5389 (0.0107)	54	0.5947 (0.0028)	6	0.4061 (0.0014)	6	0.6271 (0.0023)	36
r-AdjNorm	0.5722 (0.0134)	108	0.6742 (0.0047)	72	0.4590 (0.0024)	12	0.6083 (0.0031)	12
NAVIP	0.5664 (0.0163)	54	0.6701 (0.0039)	36	0.4565 (0.0017)	6	0.6186 (0.0024)	6
DR-LightGCN	0.5724 (0.0205)	54	0.5528 (0.0091)	6	–	–	0.6212 (0.0071)	6
SimGCL	0.5528 (0.0141)	108	0.6348 (0.0048)	12	0.4196 (0.0022)	12	0.5637 (0.0031)	12
MACR	0.5418 (0.0134)	36	0.6221 (0.0038)	18	0.4311 (0.0019)	12	0.5816 (0.0033)	12
AdvInfoNCE-LGN	0.5222 (0.0204)	216	0.6477 (0.0048)	16	0.4480 (0.0017)	8	0.5776 (0.0036)	8
DR-JL	0.4811 (0.0164)	18	0.5225 (0.0043)	9	0.3759 (0.0021)	6	0.6332 (0.0026)	6
MRDR	0.4869 (0.0187)	18	0.5248 (0.0059)	9	–	–	0.6328 (0.0020)	6
iALS	0.5812 (0.0101)	24	0.6625 (0.0043)	12	0.4528 (0.0017)	12	0.6133 (0.0027)	12
iALS (grid B)	0.5869 (0.0118)	18	–	–	–	–	–	–
iALS-p (add prior)	0.5984 (0.0121)	90	–	–	–	–	0.6273 (0.0016)	8
LightGCN-p (add prior)	0.6020 (0.0148)	108	0.6690 (0.0038)	24	0.4565 (0.0029)	12	0.6288 (0.0020)	24
DR-iALS	0.5871 (0.0113)	48	–	–	–	–	–	–
DR-BPR-LGN	0.5912 (0.0122)	108	–	–	–	–	–	–
DR-BPR-LGN-1 (Q-only)	0.5887 (0.0132)	54	–	–	–	–	–	–
DR-BPR-LGN-1 (Q-only)	–	–	0.6837 (0.0057)	36	–	–	–	–
DR-BPR-LGN-p	0.6039 (0.0139)	108	–	–	–	–	–	–
DR-BPR-LGN-p2	0.6046 (0.0177)	108	–	–	–	–	–	–
QP family (pooled)	0.6015 (0.0162)	324	0.6866 (0.0034)	84	0.4613 (0.0024)	30	0.6285 (0.0022)	120
control: shuffled Q	0.5689 (0.0148)	54	0.6689 (0.0054)	12	–	–	–	–
control: popularity Q	0.5751 (0.0118)	54	0.6550 (0.0046)	12	–	–	–	–

Pooled rows (QP family) pool the configurations of their jobs; a cell marked – has no run for that method and dataset.

Table 5: Sensitivity of the primary conclusions to evaluation choices. A check mark means the sign of the difference (and, for Main, significance at $p_{\text{Holm}} < 0.05$) survives the choice; the value is the difference $\times 10^3$ under that choice. Main: per-user difference. Oracle, configuration-oracle and matched-budget columns: split-mean difference, compared in sign with the split-mean validation-selected difference.

Dataset	Fam.	Comparison	Main (Holm)	Test oracle	Config. oracle	Matched k	$k_{\min}$	Flip share	Seeds	NB	Alt. metric
Coat	P1	DR-LightGCN – LightGCN	× -0.65	✓ +1.23	× -6.63	✓ +4.11	54	0.0%	n/a	×	×
Coat	P2	iALS-p (add prior) – iALS	× +16.82	✓ +31.66	✓ +21.23	✓ +16.12	24	0.0%	n/a	×	✓
Coat	P2	LightGCN-p (add prior) – LightGCN	✓ +31.98	✓ +27.74	✓ +28.27	✓ +34.12	54	0.0%	n/a	×	✓
Coat	P3	QP family (pooled) – LightGCN-p (add prior)	× +0.05	× +2.17	× +1.42	✓ -0.36	108	35.3%	n/a	×	×
Coat	P3	DR-iALS – iALS-p (add prior)	× -10.22	✓ -24.87	✓ -13.87	✓ -11.21	48	0.0%	n/a	×	✓
Yahoo	P1	r-AdjNorm – LightGCN	✓ +4.10	✓ +2.35	✓ +0.57	✓ +2.87	36	2.4%	✓	×	✓
Yahoo	P2	LightGCN-p (add prior) – LightGCN	× -1.43	✓ -0.07	✓ -1.11	✓ -2.16	24	0.0%	n/a	×	✓
KuaiRand	P1	r-AdjNorm – LightGCN	✓ +1.99	✓ +1.76	✓ +1.80	✓ +1.23	6	9.4%	✓	×	✓
KuaiRand	P2	LightGCN-p (add prior) – LightGCN	× -0.81	✓ -0.03	✓ -0.84	✓ -0.79	6	7.1%	n/a	×	✓
KuaiRec	P1	DR-LightGCN – LightGCN	✓ +13.17	✓ +14.69	✓ +12.44	✓ +12.97	6	0.0%	n/a	×	×
KuaiRec	P2	LightGCN-p (add prior) – LightGCN	✓ +21.22	✓ +20.02	✓ +20.00	✓ +20.25	24	0.0%	n/a	✓	✓
KuaiRec	P2	iALS-p (add prior) – iALS	✓ +14.74	✓ +14.00	✓ +14.00	✓ +16.89	8	0.0%	n/a	✓	✓
KuaiRec	P3	QP family (pooled) – LightGCN-p (add prior)	× -0.32	× +0.32	× +0.46	✓ -0.12	24	30.9%	n/a	×	✓
KuaiRec	P4	DRUP – DR-JL	✓ +4.38	✓ +5.48	✓ +5.45	× -14.90	6	78.4%	n/a	×	✓

Flip share: fraction of 1,000 random subgrids of $k_{\min}$ configurations per method whose sign differs from the full grid. Alt. metric: nDCG@10 (Coat, Yahoo), nDCG@20 (KuaiRand), nDCG@50 (KuaiRec), on the stored selections.

Table 6: Primary comparisons P1–P4 (Section 3): paired per-user differences, tests, MDE, TOST at the sensitivity margins (Section 3.6), and the matched-budget difference at $k = k_{\min}$.

Dataset	Fam.	Comparison ($A - B$)	Diff. $\times 10^3$ [95% CI]	p	p_{Holm}	$p_{\text{NB,Holm}}$	MDE $\times 10^3$	TOST at margin	Matched-k diff $\times 10^3$	Seeds 1–3 diff $\times 10^3$
Coat	P1	DR-LightGCN – LightGCN	-0.65 [30.16]	0.966	1.000	1.000	43.07	does not exclude $\lvert d\rvert > 0.010$ (p=0.271)	+4.11	–
Coat	P2	iALS-p (add prior) – iALS	+16.82 [18.04]	0.068	0.270	1.000	25.76	does not exclude $\lvert d\rvert > 0.010$ (p=0.771)	+16.12	–
Coat	P2	LightGCN-p (add prior) – LightGCN	+31.98 [20.65]	0.003	0.013	1.000	29.48	does not exclude $\lvert d\rvert > 0.010$ (p=0.981)	+34.12	–
Coat	P3	QP family (pooled) – LightGCN-p (add prior)	+0.05 [8.29]	0.991	1.000	1.000	11.83	excludes $\lvert d\rvert > 0.010$ (p=0.009)	-0.36	–
Coat	P3	DR-iALS – iALS-p (add prior)	-10.22 [15.96]	0.208	0.625	1.000	22.79	does not exclude $\lvert d\rvert > 0.010$ (p=0.511)	-11.21	–
Yahoo	P1	r-AdjNorm – LightGCN	+4.10 [3.29]	0.015	0.029	0.883	4.70	yes (p=<0.001)	+2.87	+5.26 (p=<0.001)
Yahoo	P2	LightGCN-p (add prior) – LightGCN	-1.43 [3.93]	0.475	0.475	0.883	5.62	yes (p=<0.001)	-2.16	–
KuaiRand	P1	r-AdjNorm – LightGCN	+1.99 [1.02]	<0.001	<0.001	1.000	1.46	yes (p=0.026)	+1.23	+2.56 (p=<0.001)
KuaiRand	P2	LightGCN-p (add prior) – LightGCN	-0.81 [1.33]	0.234	0.234	1.000	1.90	yes (p=<0.001)	-0.79	–
KuaiRec	P1	DR-LightGCN – LightGCN	+13.17 [3.31]	<0.001	<0.001	0.787	4.74	no (p=1.000)	+12.97	–
KuaiRec	P2	LightGCN-p (add prior) – LightGCN	+21.22 [2.86]	<0.001	<0.001	0.009	4.09	no (p=1.000)	+20.25	–
KuaiRec	P2	iALS-p (add prior) – iALS	+14.74 [3.43]	<0.001	<0.001	0.043	4.90	no (p=1.000)	+16.89	–
KuaiRec	P3	QP family (pooled) – LightGCN-p (add prior)	-0.32 [0.50]	0.211	0.211	0.871	0.72	yes (p=<0.001)	-0.12	–
KuaiRec	P4	DRUP – DR-JL	+4.38 [2.21]	<0.001	<0.001	0.426	3.15	no (p=0.890)	-14.90	–

Margins are sensitivity thresholds chosen before the analysis and not derived from the literature (Section 3.6). On Coat the TOST column states the differences the data exclude; equivalence is not claimed there. Seeds column: only comparisons whose baselines were retrained with seeds 1–3 (Section 3.5) are shown.

Table 7: Prior anatomy. Test nDCG of iALS-p and LightGCN-p under four additive-imputation variants, and of the prior alone on all four datasets.

Dataset	Learner / prior	nDCG (mean ± sd)	Δ vs no prior $\times 10^3$ [CI]	p_{Holm}	Δ vs add $\times 10^3$ [CI]	p_{Holm}
Coat	iALS (no prior)	0.5812 ± 0.0107	–	–	–	–
Coat	iALS-p, add (IPS-fitted)	0.5984 ± 0.0128	+16.82 [18.04]	0.811	–	–
Coat	iALS-p, add, no IPS	0.6044 ± 0.0115	+23.08 [18.24]	0.187	+6.27 [6.97]	0.856
Coat	iALS-p, pop	0.5832 ± 0.0238	+1.67 [16.11]	1.000	-15.15 [15.39]	0.698
Coat	iALS-p, shuffled item bias	0.5683 ± 0.0175	-14.06 [18.53]	1.000	-30.87 [22.75]	0.120
Coat	LightGCN (no prior)	0.5683 ± 0.0139	–	–	–	–
Coat	LightGCN-p, add (IPS-fitted)	0.6020 ± 0.0156	+31.98 [20.65]	0.046	–	–
Coat	LightGCN-p, add, no IPS	0.6052 ± 0.0145	+36.00 [16.88]	<0.001	+4.01 [10.01]	1.000
Coat	LightGCN-p, pop	0.5835 ± 0.0233	+15.63 [10.68]	0.069	-16.36 [19.52]	1.000
Coat	LightGCN-p, shuffled item bias	0.5527 ± 0.0095	-12.95 [24.18]	1.000	-44.93 [29.53]	0.051
KuaiRec	iALS (no prior)	0.6133 ± 0.0030	–	–	–	–
KuaiRec	iALS-p, add (IPS-fitted)	0.6273 ± 0.0018	+14.74 [3.43]	<0.001	–	–
KuaiRec	iALS-p, add, no IPS	0.6350 ± 0.0021	+22.79 [4.04]	<0.001	+8.05 [2.06]	<0.001
KuaiRec	iALS-p, pop	0.6144 ± 0.0028	+1.06 [2.46]	1.000	-13.68 [3.31]	<0.001
KuaiRec	iALS-p, shuffled item bias	0.4993 ± 0.0032	-112.07 [9.39]	<0.001	-126.82 [9.29]	<0.001
KuaiRec	LightGCN (no prior)	0.6085 ± 0.0031	–	–	–	–
KuaiRec	LightGCN-p, add (IPS-fitted)	0.6288 ± 0.0023	+21.22 [2.86]	<0.001	–	–
Coat	prior alone, add (IPS-fitted)	0.5585 ± 0.0105	–	–	–	–
Coat	prior alone, add, no IPS	0.5574 ± 0.0107	–	–	-2.09 [8.46]	1.000
Coat	prior alone, pop	0.5367 ± 0.0100	–	–	-19.40 [27.79]	1.000
Coat	prior alone, shuffled item bias	0.3572 ± 0.0125	–	–	-203.62 [51.90]	<0.001
Coat	prior alone, user only (b_i = 0)	0.3497 ± 0.0152	–	–	-207.15 [49.91]	<0.001
Yahoo	prior alone, add (IPS-fitted)	0.5243 ± 0.0050	–	–	–	–
Yahoo	prior alone, add, no IPS	0.5237 ± 0.0055	–	–	-1.09 [3.98]	1.000
Yahoo	prior alone, pop	0.5331 ± 0.0065	–	–	+8.90 [14.93]	1.000
Yahoo	prior alone, shuffled item bias	0.3446 ± 0.0056	–	–	-177.98 [17.37]	<0.001
Yahoo	prior alone, user only (b_i = 0)	0.3767 ± 0.0051	–	–	-144.74 [17.38]	<0.001
KuaiRand	prior alone, add (IPS-fitted)	0.3637 ± 0.0013	–	–	–	–
KuaiRand	prior alone, add, no IPS	0.3837 ± 0.0023	–	–	+19.90 [2.45]	<0.001
KuaiRand	prior alone, pop	0.4204 ± 0.0021	–	–	+56.41 [5.19]	<0.001
KuaiRand	prior alone, shuffled item bias	0.3257 ± 0.0018	–	–	-39.12 [5.64]	<0.001
KuaiRand	prior alone, user only (b_i = 0)	0.3259 ± 0.0011	–	–	-38.84 [5.53]	<0.001
KuaiRec	prior alone, add (IPS-fitted)	0.6216 ± 0.0016	–	–	–	–
KuaiRec	prior alone, add, no IPS	0.6367 ± 0.0022	–	–	+15.16 [1.68]	<0.001
KuaiRec	prior alone, pop	0.5766 ± 0.0026	–	–	-45.70 [4.25]	<0.001
KuaiRec	prior alone, shuffled item bias	0.0398 ± 0.0007	–	–	-582.01 [10.19]	<0.001
KuaiRec	prior alone, user only (b_i = 0)	0.0686 ± 0.0010	–	–	-554.01 [10.16]	<0.001

Δ = per-user paired difference (mean over splits), $\times 10^3$, [half-width of the 95% CI]. No-prior references: stored iALS and LightGCN. The add row is the IPS-fitted additive prior. p_{Holm} over all paired tests of this table. user only (b_i = 0) gives the same score to every item for a given user, so its ranking is decided by tie-breaking. Coat LightGCN-p and iALS-p use 10 splits, KuaiRec 5. Rows absent from the table were not run (only the prior-alone rows exist for those priors).

Table 8: Refinement family against the simple debiased prior of the same learner, and the shuffled and popularity controls against the backbone.

Dataset	Family	Method	Mean	Cfg	Versus	Diff. $\times 10^3$ [95% CI]	p_{Holm}	TOST (Coat: exclusion, not equivalence)
Coat	refinement vs simple prior	DR-BPR-LGN	0.5912	108	LightGCN-p (add prior)	-8.02 [18.85]	1.000	does not exclude
Coat	refinement vs simple prior	DR-BPR-LGN-1 (Q-only)	0.5887	54	LightGCN-p (add prior)	-10.79 [19.26]	1.000	does not exclude
Coat	refinement vs simple prior	DR-BPR-LGN-p	0.6039	108	LightGCN-p (add prior)	+2.09 [6.48]	1.000	excludes $\|d\| > 0.010$
Coat	refinement vs simple prior	DR-BPR-LGN-p2	0.6046	108	LightGCN-p (add prior)	+3.27 [6.94]	1.000	excludes $\|d\| > 0.010$
Coat	refinement vs simple prior	QP family (pooled)	0.6015	324	LightGCN-p (add prior)	+0.05 [8.29]	1.000	excludes $\|d\| > 0.010$
Coat	refinement vs simple prior (iALS)	DR-iALS	0.5871	48	iALS-p (add prior)	-10.22 [15.96]	0.208	does not exclude
Coat	controls vs backbone	control: shuffled Q	0.5689	54	LightGCN	+0.97 [5.50]	0.730	excludes $\|d\| > 0.010$
Coat	controls vs backbone	control: popularity Q	0.5751	54	LightGCN	+6.91 [6.04]	0.050	does not exclude
Yahoo	refinement vs simple prior	DR-BPR-LGN-1 (Q-only)	0.6837	36	LightGCN-p (add prior)	+15.15 [5.84]	<0.001	no
Yahoo	refinement vs simple prior	QP family (pooled)	0.6866	84	LightGCN-p (add prior)	+18.20 [5.64]	<0.001	no
Yahoo	controls vs backbone	control: shuffled Q	0.6689	12	LightGCN	-1.01 [3.20]	0.534	yes
Yahoo	controls vs backbone	control: popularity Q	0.6550	12	LightGCN	-15.91 [6.06]	<0.001	no
KuaiRand	refinement vs simple prior	QP family (pooled)	0.4613	30	LightGCN-p (add prior)	+4.74 [1.54]	<0.001	no
KuaiRec	refinement vs simple prior	QP family (pooled)	0.6285	120	LightGCN-p (add prior)	-0.32 [0.50]	0.211	yes

Holm within each family and dataset. Rows are missing where the simple prior has not been run for that dataset.

Table 9: Coat with estimated propensities versus the shipped propensities.

Method	Propensities	nDCG (mean ± sd)	Δ vs no-prior $\times 10^3$ [CI]	p_{Holm}	Estimated − shipped $\times 10^3$ [CI]	p_{Holm}
iALS-p	shipped	0.5984 ± 0.0128	+16.82 [18.04]	0.380	–	–
iALS-p	estimated	0.6009 ± 0.0120	+19.44 [18.66]	0.330	+2.63 [5.27]	1.000
LightGCN-p	shipped	0.6020 ± 0.0156	+31.98 [20.65]	0.023	–	–
LightGCN-p	estimated	0.6025 ± 0.0154	+33.62 [16.72]	<0.001	+1.64 [10.30]	1.000
iALS (grid B), sanity check	shipped	0.5869 ± 0.0124	+7.39 [7.52]	0.380	–	–
iALS (grid B), sanity check	estimated	0.5869 ± 0.0124	+7.39 [7.52]	0.380	0 (identical)	1
imputation alone (add)	shipped	0.5585 ± 0.0105	–	–	–	–
imputation alone (add)	estimated	0.5573 ± 0.0097	–	–	-1.14 [7.26]	1.000
iALS (no prior)	shipped	0.5812 ± 0.0107	–	–	–	–
LightGCN (no prior)	shipped	0.5683 ± 0.0139	–	–	–	–

The no-prior reference of each learner is the stored run. Plain iALS and LightGCN do not read the propensities (only NAVIP does), so the shipped no-prior run is the reference for the estimated rows. LightGCN with estimated propensities was not run. The iALS (grid B) row is a sanity check: plain iALS does not read the propensities, so its difference is zero by construction. Only the prior rows and the imputation alone were refitted. p_{Holm} over the tests of this table.

Table 10: Coat iALS with the wider grid against the stored iALS and iALS-p.

Model	Grid	Cfg	nDCG (mean ± sd)	Δ vs iALS $\times 10^3$ [CI]	p_{Holm}	Δ vs iALS-p $\times 10^3$ [CI]	p_{Holm}	Selected values (number of splits)
iALS (grid A, stored)	alpha {3,10,30} x reg {1,10,100,1000} x dim {32,64}	24	0.5812 ± 0.0107	ref.	–	–	–	alpha 30:10; reg 100:10
iALS (grid B)	alpha {10,30,100} x reg {10,100,300} x dim {32,64}	18	0.5869 ± 0.0124	–	–	–	–	alpha 30:1, 100:9; reg 100:4, 300:6
iALS (grid C)	alpha {30,100,300,1000} x reg {10,100,300,1000} x dim {32,64}	32	0.5922 ± 0.0150	+12.92 [8.01]	0.003	-3.90 [15.99]	0.631	alpha 30:1, 100:4, 300:1, 1000:4; reg 100:2, 300:3, 1000:5
iALS-p (add, stored grid)	alpha {10,30,100} x reg {10,100,300} x pbeta {0.1,0.2,0.3,0.5,1} x dim {32,64}	90	0.5984 ± 0.0128	–	–	–	–	alpha 30:2, 100:8; reg 100:4, 300:6

Δ = per-user paired difference (mean over the 10 splits), $\times 10^3$, [half-width of the 95% CI]. p_{Holm} over the two comparisons of grid C. Selected values are counted over the 10 splits. Grid C was widened after grid B and is not exhausted (alpha 1000 is chosen in 4 splits).

Table 11: Walk correction at matched configurations (Coat, Yahoo), and KuaiRec DRUP minus DR.

Dataset	β	Mean diff. $\times 10^3$	Share of configs < 0	Splits with mean < 0
Coat	0.03	-0.33	0.49	7/10
Coat	0.1	-1.28	0.52	9/10
Coat	0.3	-2.92	0.54	10/10
Coat	1	-5.09	0.74	10/10
Coat	3	-7.35	0.74	10/10
Coat	10	-7.44	0.80	10/10
Coat	1000	-8.17	0.90	10/10
Yahoo	0.03	-0.02	0.41	5/5
Yahoo	0.1	-0.03	0.50	4/5
Yahoo	0.3	-0.12	0.56	5/5
Yahoo	1	-0.04	0.47	3/5
Yahoo	3	-0.07	0.28	4/5
Yahoo	10	-0.19	0.41	5/5
Yahoo	1000	-0.32	0.38	5/5
KuaiRec (DRUP − DR, all splits)	–	-0.12	–	–

KuaiRec row: paired test p=0.113; within the sensitivity margin ± 0.003 (Section 3.6).

Table A.12: Appendix: every method against LightGCN on Coat (backbone family, nDCG@5). Holm is taken over this family.

Method	Diff. $\times 10^3$ [95% CI]	p	p_{Holm}	$p_{\text{NB,Holm}}$	MDE $\times 10^3$
MF	-93.59 [33.10]	<0.001	<0.001	0.413	47.26
IPS-MF	-98.45 [33.78]	<0.001	<0.001	0.196	48.23
DR-MF	-47.85 [32.93]	0.005	0.106	0.196	47.02
BPR-MF	-53.28 [21.08]	<0.001	<0.001	1.000	30.10
PDA	-50.81 [20.12]	<0.001	<0.001	1.000	28.73
LightGCN-pt	-32.48 [32.29]	0.049	0.779	1.000	46.11
r-AdjNorm	+3.76 [5.74]	0.198	1.000	1.000	8.20
NAVIP	-1.59 [10.13]	0.758	1.000	1.000	14.47
DR-LightGCN	-0.65 [30.16]	0.966	1.000	1.000	43.07
DR-JL	-91.00 [35.14]	<0.001	<0.001	0.203	50.18
MRDR	-85.16 [36.17]	<0.001	<0.001	0.734	51.65
MACR	-29.43 [24.60]	0.019	0.405	1.000	35.13
SimGCL	-17.49 [23.88]	0.150	1.000	1.000	34.09
iALS	+12.01 [16.12]	0.143	1.000	1.000	23.01
AdvInfoNCE-LGN	-46.47 [25.16]	<0.001	0.009	1.000	35.93
iALS (grid B)	+19.40 [16.48]	0.021	0.425	1.000	23.53
iALS-p (add prior)	+28.83 [20.48]	0.006	0.132	1.000	29.25
LightGCN-p (add prior)	+31.98 [20.65]	0.003	0.061	1.000	29.48
DR-BPR-LGN	+23.96 [8.67]	<0.001	<0.001	1.000	12.38
DR-BPR-LGN-1 (Q-only)	+21.20 [9.26]	<0.001	<0.001	1.000	13.22
DR-BPR-LGN-p	+34.07 [20.11]	<0.001	0.025	1.000	28.72
DR-BPR-LGN-p2	+35.26 [20.51]	<0.001	0.022	1.000	29.28
DR-iALS	+18.61 [16.31]	0.026	0.480	1.000	23.28
QP family (pooled)	+32.03 [17.27]	<0.001	0.009	1.000	24.65
control: shuffled Q	+0.97 [5.50]	0.730	1.000	1.000	7.86
control: popularity Q	+6.91 [6.04]	0.025	0.480	1.000	8.63
Obs	-12.94 [30.72]	0.408	1.000	1.000	43.87
EASE	-28.72 [32.13]	0.080	1.000	1.000	45.88
GF-CF	-18.32 [30.12]	0.232	1.000	1.000	43.01
IPS	-18.36 [31.48]	0.252	1.000	1.000	44.96
DR	-26.62 [32.00]	0.103	1.000	1.000	45.69
DRUP	-25.21 [32.81]	0.131	1.000	1.000	46.85
EASE-DR	-15.27 [33.69]	0.373	1.000	1.000	48.10
GF-CF-DR	-12.33 [32.46]	0.455	1.000	1.000	46.35
Pop	-34.67 [31.12]	0.029	0.495	1.000	44.43
Impute	-16.97 [34.67]	0.336	1.000	1.000	49.50

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.6.

Table A.13: Appendix: every method against LightGCN on Yahoo (backbone family, nDCG@5). Holm is taken over this family.

Method	Diff. $\times 10^3$ [95% CI]	p	p_{Holm}	$p_{\text{NB,Holm}}$	MDE $\times 10^3$
MF	-162.69 [15.25]	<0.001	<0.001	0.021	21.80
IPS-MF	-159.11 [15.20]	<0.001	<0.001	0.012	21.72
DR-MF	-162.10 [16.02]	<0.001	<0.001	0.012	22.89
BPR-MF	-26.53 [8.17]	<0.001	<0.001	0.218	11.68
PDA	-20.36 [7.73]	<0.001	<0.001	0.021	11.05
LightGCN-pt	-76.19 [13.05]	<0.001	<0.001	0.061	18.65
r-AdjNorm	+4.10 [3.29]	0.015	0.073	1.000	4.70
NAVIP	-0.74 [3.11]	0.642	1.000	1.000	4.45
DR-LightGCN	-117.88 [14.56]	<0.001	<0.001	0.019	20.81
DR-JL	-149.61 [14.97]	<0.001	<0.001	0.016	21.39
MRDR	-146.84 [14.97]	<0.001	<0.001	0.006	21.40
MACR	-48.31 [9.71]	<0.001	<0.001	0.021	13.87
SimGCL	-36.49 [9.61]	<0.001	<0.001	0.042	13.74
iALS	-8.11 [7.99]	0.047	0.187	1.000	11.42
AdvInfoNCE-LGN	-21.08 [7.78]	<0.001	<0.001	0.135	11.12
LightGCN-p (add prior)	-1.43 [3.93]	0.475	1.000	1.000	5.62
DR-BPR-LGN-1 (Q-only)	+13.72 [5.27]	<0.001	<0.001	0.291	7.54
QP family (pooled)	+16.77 [5.82]	<0.001	<0.001	0.434	8.31
control: shuffled Q	-1.01 [3.20]	0.534	1.000	1.000	4.57
control: popularity Q	-15.91 [6.06]	<0.001	<0.001	1.000	8.66
Obs	-12.40 [8.49]	0.004	0.025	1.000	12.13
DR	-14.58 [8.51]	<0.001	0.006	0.511	12.16
DRUP	-14.58 [8.51]	<0.001	0.006	0.511	12.16
Pop	-138.65 [13.06]	<0.001	<0.001	<0.001	18.67
Impute	-155.70 [16.32]	<0.001	<0.001	0.016	23.32

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.6.

Table A.14: Appendix: every method against LightGCN on KuaiRand (backbone family, nDCG@10). Holm is taken over this family.

Method	Diff. $\times 10^3$ [95% CI]	p	p_{Holm}	$p_{\text{NB,Holm}}$	MDE $\times 10^3$
MF	-87.35 [5.38]	<0.001	<0.001	<0.001	7.69
DR-MF	-96.98 [5.32]	<0.001	<0.001	0.010	7.61
BPR-MF	-10.06 [2.62]	<0.001	<0.001	0.574	3.74
LightGCN-pt	-52.46 [4.85]	<0.001	<0.001	0.001	6.93
r-AdjNorm	+1.99 [1.02]	<0.001	<0.001	1.000	1.46
NAVIP	-0.43 [1.14]	0.461	0.468	1.000	1.63
DR-JL	-82.06 [4.90]	<0.001	<0.001	<0.001	7.01
MACR	-26.46 [3.58]	<0.001	<0.001	0.012	5.12
SimGCL	-38.27 [4.12]	<0.001	<0.001	0.005	5.89
iALS	-5.05 [3.16]	0.002	0.005	1.000	4.51
AdvInfoNCE-LGN	-8.86 [2.50]	<0.001	<0.001	0.120	3.57
LightGCN-p (add prior)	-0.81 [1.33]	0.234	0.468	1.000	1.90
QP family (pooled)	+3.93 [1.53]	<0.001	<0.001	1.000	2.19

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.6.

Table A.15: Appendix: every method against LightGCN on KuaiRec (backbone family, nDCG@20). Holm is taken over this family.

Method	Diff. $\times 10^3$ [95% CI]	p	p_{Holm}	$p_{\text{NB,Holm}}$	MDE $\times 10^3$
LightGCN-pt	+19.59 [4.55]	<0.001	<0.001	0.030	6.51
r-AdjNorm	+0.22 [0.80]	0.585	0.585	0.877	1.14
NAVIP	+10.41 [1.82]	<0.001	<0.001	0.034	2.60
DR-LightGCN	+13.17 [3.31]	<0.001	<0.001	0.787	4.74
BPR-MF	-11.73 [3.27]	<0.001	<0.001	0.057	4.68
SimGCL	-44.34 [4.11]	<0.001	<0.001	<0.001	5.88
DR-JL	+25.46 [3.66]	<0.001	<0.001	0.057	5.22
MRDR	+25.24 [3.53]	<0.001	<0.001	0.031	5.05
MACR	-25.84 [4.57]	<0.001	<0.001	0.018	6.53
iALS	+4.98 [2.38]	<0.001	<0.001	0.107	3.41
MF	+14.63 [3.96]	<0.001	<0.001	0.353	5.65
IPS-MF	+17.69 [3.53]	<0.001	<0.001	0.089	5.04
DR-MF	+14.47 [4.20]	<0.001	<0.001	0.124	6.00
AdvInfoNCE-LGN	-28.42 [6.46]	<0.001	<0.001	0.034	9.23
LightGCN-p (add prior)	+21.22 [2.86]	<0.001	<0.001	0.030	4.09
iALS-p (add prior)	+19.72 [3.32]	<0.001	<0.001	0.058	4.74
iALS-p (add, DRGP grid)	+19.96 [2.71]	<0.001	<0.001	0.016	3.87
QP family (pooled)	+20.90 [2.88]	<0.001	<0.001	0.089	4.11
Obs	-4.80 [3.11]	0.002	0.005	0.094	4.44
DRUP	+29.84 [3.51]	<0.001	<0.001	0.011	5.01
DR	+29.96 [3.51]	<0.001	<0.001	0.011	5.01
Pop	-31.41 [3.25]	<0.001	<0.001	<0.001	4.64
Impute	+11.89 [3.93]	<0.001	<0.001	0.301	5.62

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.6.